

The Pre-Opening Period, High-Frequency Trading, and Market Liquidity

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ABSTRACT

We use order-level data from the Tokyo Stock Exchange with anonymous trader identifiers to document that 20% of high-frequency traders (HFTs) participate in the pre-opening period (Active HFTs). Their participation in the pre-opening period is associated with improved market liquidity. Active HFTs are largely market-makers, contributing to price discovery through quotes rather than trades, supplying liquidity for two-thirds of their trading volume, and are at the top of the book 1.6 times as often as Non-Active HFTs. We find no support for the information channel behind the participation decision; rather, it reflects the benefit of securing a superior queue position.

JEL classification: G12, G14.

Keywords: High-frequency traders (HFTs); Pre-opening period; Information; Queue position; Opportunistic traders; Market-Makers

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Rapid advances in trading technology over the past two decades have fundamentally altered the structure of global equity markets, reshaping the behavior of liquidity providers and consumers alike and prompting intense regulatory and academic scrutiny of high-frequency traders (HFTs).¹ Most equity markets comprise three distinct phases: a pre-opening period, during which quotes are submitted but no transactions occur; an opening call auction, in which buy and sell orders are crossed to determine the opening price; and a continuous trading period, when normal trading activity occurs, with posted quotes and trades. The pre-opening period is of particular interest since it represents the earliest point at which information accumulated overnight is impounded into market prices, and because the absence of transactions neutralizes the speed advantage – the defining characteristic of HFTs. A priori, it is not obvious that any type of HFTs should be present during the pre-opening period, since the absence of executions removes the immediacy motive of opportunistic traders and the opportunity to earn a spread for market-makers. Yet, HFTs do participate, and they do so selectively. On a typical stock-day in our sample, only around 20% of HFT desks submit orders before the market opens. In this paper, we investigate which type of HFT (opportunistic vs. market-making) is present in the pre-opening period, the theoretical predictions in the previous literature for why they are, and whether their presence is associated with improved or deteriorated market liquidity during the continuous trading period that follows.

Answering these questions requires the ability to track individual traders across the pre-opening period, the opening call auction, and the continuous trading period. Existing papers on HFTs and the pre-opening period (see [Anagnostidis, Fontaine, and Varsakelis \(2020\)](#), [Bellia, Pelizzon, Subrahmanyam, Uno, and Yuferova \(2017\)](#), and [Daures, Moinas, and Boussetta \(2025\)](#)) treat HFTs as a single homogeneous group due to data limitations and are not able to distinguish HFTs that participate in the pre-opening period from those that do not. These limitations are binding for the question at hand, since the variation of interest lies precisely within the heterogeneity of HFTs: some traders choose to be present before the market

¹See [Lewis \(2014\)](#) for a popular account. We refer to [Jones \(2013\)](#), [Biais and Foucault \(2014\)](#), and [Menkveld \(2016\)](#) for a more balanced assessment of the implications of HFT for market quality.

opens while others do not, and the same trader makes one choice on some stock-days and the opposite choice on others.

We address this gap using a unique dataset provided by the Tokyo Stock Exchange (TSE), one of the largest stock markets in the world, which also has a large presence of HFTs.² For our analysis, we focus on the constituents of the TOPIX100, one of the main Japanese stock market indices.³

By investigating order and trade data with anonymous trader identifiers obtained from the TSE for April 2013 for the TOPIX100 index constituents, we identify 13 HFTs using criteria based on speed and end-of-day inventory position throughout our sample period. Separately, we classify each HFT as Active or Non-Active on a stock-day basis, depending on whether it submits orders during the morning pre-opening period. While HFT status is fixed for a given trader throughout the sample, Active status varies from one stock-day to another, so that our tests exploit variation within the set of HFTs. Throughout, we emphasize that our empirical design is associational: we document patterns in the data and assess whether they are consistent with particular channels, but we do not claim to identify causal effects.

Our first set of results establishes which type of HFT chooses to be present before the market opens. The literature distinguishes market-making from opportunistic HFTs in the continuous trading period (SEC (2010), Hagströmer and Norden (2013)), and we map this taxonomy onto the participation decision using three independent measures, all of which point in the same direction. First, using the structural vector-autoregression (SVAR) ap-

²Hosaka (2014) documents that HFT activity amounted to 25% of traded value as of January 2013 on the TSE, while ESMA (2014) documents 24% of traded value on the European markets as of May 2013, as stemming from HFTs. In 2020, HFTs represented about 50% of trading volume in U.S. equity markets. In European equity markets, their share has been estimated to be between 24% and 43% of trading volume and about 58% to 76% of orders (Breckenfelder (2020)). More recent estimates suggest the numbers are even higher today.

³Our analysis is not affected by information flows of stocks traded in multiple locations, whether alternative trading venues or other stock exchanges. We note that the TSE’s market share of all Japanese stocks is over 90%; in addition, the TSE is in a significantly different time zone than the major stock exchanges in Europe and the U.S. (which trade some Japanese stocks directly or in the form of depositary receipts) and hence opens the trading day with a completely different information set.

proach (following [Hasbrouck \(1991\)](#) and [Brogaard, Hendershott, and Riordan \(2019\)](#)), we find that Non-Active HFTs contribute to price discovery predominantly through their trades, whose price impact is 1.4 times larger than that of trades initiated by Active HFTs (6.261 bps vs. 4.434 bps), while Active HFTs contribute to price discovery through quote adjustments, without consuming liquidity (1.282 bps vs. 0.555 bps). Second, Non-Active HFTs are neutral with respect to trading volume imbalance between liquidity-providing and liquidity-consuming transactions, whereas Active HFTs largely supply liquidity in their transactions with a trading volume imbalance ratio of around 67%. Third, Active HFTs are present at the top and at the best bid-offer level of the limit order book 1.6 and 1.2 times more often than Non-Active HFTs, with the difference disappearing at the three best levels, where time priority is of a lesser concern. Taken together, these results indicate that Active HFTs are largely market-makers, while Non-Active HFTs are opportunistic traders.

Having established which type of HFTs chooses to be present before the market opens, we turn to the question of why. An information channel is a natural potential candidate. For example, [Medrano and Vives \(2001\)](#) show that a strategic informed trader might choose to come to the pre-opening period and try to manipulate the opening price in her favor. Even though this channel is more relevant for opportunistic HFTs, we test it to reinforce our claim of Active HFTs being market-makers.

If the information channel is indeed behind the decision to come to the pre-opening period, the likelihood of participation should increase with information intensity. The evidence, however, does not support it. Participation is not more likely when the overnight information environment is intense; if anything, the association runs in the opposite direction, with the absolute overnight return associated negatively with the likelihood of participation in the pre-opening period. Overnight earnings announcements, prior-day volatility, and return reversals do not determine the decision of HFT to come to the pre-opening period either. In addition, if HFTs are able to gather valuable information from the pre-opening period it should be reflected in their profits. However, profits of Active and Non-Active HFTs are statistically indistinguishable. For attempts of manipulation, one would also expect that

the price impact of messages submitted by HFTs during the pre-opening period to be larger than that of messages submitted by Non-HFTs during the pre-opening period. In unreported results, we show that the price impact of aggressive messages from Active Non-HFTs during the pre-opening period (2.702 bps) is considerably larger than that of Active HFTs (0.233 bps), which does not support this explanation. Overall, the information channel does not find support in the empirical tests we conduct.

The more likely explanation for the participation decision is the liquidity provision channel, namely, securing queue position (e.g., [Menkveld and Zoican \(2017\)](#) model market-making vs. opportunistic HFTs as a deterministic function of who is the first to submit the order at the best bid-offer level). Any order left unexecuted at the opening auction is automatically carried into the continuous trading period with time priority superior to that of orders entered afterwards, so that the pre-opening period offers a natural opportunity to secure a place in the limit order queue. Consistent with this interpretation, 10% of pre-opening orders from HFTs are executed during the continuous trading period, and a one standard deviation increase in that proportion is associated with an increase of 3.034% in the time presence at the best bid-offer level (which is equivalent to 9% of the sample mean).

Finally, we investigate whether the participation decision is associated with improved or deteriorated market liquidity in the continuous trading period that follows. Recomputing share-weighted liquidity measures with and without the limit orders submitted by Active HFTs, we find that their inclusion is associated with quoted spreads narrower by 0.222 bps, quoted depth greater by 0.650 mln JPY, and depth imbalance at the best bid-offer level lower by 0.076%, and that the securing queue position motive accounts for part of the depth and imbalance results.

Active HFTs are also associated with improved market liquidity during turbulent periods. Following [Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov \(2018\)](#), we identify extreme price movements (EPMs) as the 99.9th percentile of absolute 10-second midquote returns and find that, within the interval in which an EPM occurs, the net share imbalance of Active HFTs runs against the direction of the EPMs, and is driven by their

liquidity-providing orders, while Non-Active HFTs trade in the direction of the EPM through liquidity-demanding orders. Active HFTs, therefore, do not amplify extreme price movements; instead, they trade to mitigate them.

The structure of the paper is as follows. In Section 1, we survey the literature on HFTs and market quality, and on the microstructure of the pre-opening period. In Section 2, we develop our hypotheses. Section 3 describes the institutional details of the TSE, particularly for HFTs, and the special features of our database. In Section 4, we describe the procedures we use to identify HFTs, and we present our empirical analysis and results in Section 5. Some additional analysis is provided in Section 6, and Section 7 concludes.

1 Literature review

Our study contributes to two streams of literature (i) heterogeneity of HFTs in the equity markets; and (ii) the microstructure of the pre-opening period.

First, a large empirical literature documents HFTs' behavior and its consequences for market quality during the continuous trading period. The role of HFTs in price discovery and liquidity provision has been established in both normal conditions (Brogaard, Hendershott, and Riordan (2014, 2019), Malinova, Park, and Riordan (2018)) and during market stress (Kirilenko, Kyle, Samadi, and Tuzun (2017), Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov (2018), Bellia, Christensen, Kolokolov, Pelizzon, and Reno (2025)). Recent HFT-specific theoretical research deals with the speed advantage of HFTs in terms of information processing and trading. Their greater speed allows HFTs to react more quickly to public news than other traders (as in theoretical research by Cespa and Foucault (2011), Pagnotta and Philippon (2018), Ait-Sahalia and Saglam (2013), Jovanovic and Menkveld (2015), Budish, Cramton, and Shim (2015), Biais, Foucault, and Moinas (2015), and Foucault, Hombert, and Roşu (2016)). Baron, Brogaard, Hagströmer, and Kirilenko (2019) establish that relative, rather than absolute, speed drives cross-sectional differences in HFTs' profitability.

Critically for this paper, HFTs are far from homogeneous. In particular, SEC (2010)

discusses the variety of strategies employed by HFTs and categorizes them into two categories: opportunistic vs. market-making. [Hagströmer and Norden \(2013\)](#) provide empirical evidence for this categorization in the continuous trading period. The market making vs. opportunistic distinction, while well established for the continuous trading period, does not map neatly onto the pre-opening participation decision. A priori, neither of the two types has an obvious reason to be present. On the one hand, the absence of executions eliminates the immediacy motive of opportunistic HFTs and offers no direct spread-earning opportunity for market makers either. On the other hand, [Medrano and Vives \(2001\)](#) theoretically predict that a strategic informed trader can come to the pre-opening period and try to manipulate the opening price in her favor; hence, this might be a reason for opportunistic HFTs to come to the pre-opening period. Besides that, opportunistic HFTs might also use the pre-opening period for information gathering by probing the market and provoking other traders' actions by posting orders with no intention to execute them. In turn, market-making HFTs can use the pre-opening period to secure their position in the queue. For example, [Menkveld and Zoican \(2017\)](#) and [Budish, Cramton, and Shim \(2015\)](#) model both types of HFTs and show that the trader who managed to post the best quote first becomes a market maker. The importance of the queue position in the continuous trading period is also discussed by [Li, Wang, and Ye \(2021\)](#), [Goldstein, Kwan, and Philip \(2023\)](#), [Dahlström, Hagströmer, and Nordén \(2024\)](#). Any order that is not executed at the opening auction is automatically transferred to the continuous trading period (unless specified otherwise) with time priority superior to the orders entered after the opening auction takes place, thereby providing a natural opportunity to secure a superior queue position. We contribute to this literature by showing that market-making rather than opportunistic HFTs come to the pre-opening period. We also show that the information-related motive does not drive this decision, while securing queue position partially explains it.

Second, our research is also related to the microstructure of trading activity in the pre-opening period. The pattern of pre-opening market activity has been studied in the earlier literature (e.g., [Amihud and Mendelson \(1991\)](#), [Biais, Hillion, and Spatt \(1999\)](#), [Cao, Ghy-](#)

sels, and Hatheway (2000), Ciccotello and Hatheway (2000), Madhavan and Panchapagesan (2000), Barclay and Hendershott (2003), Davies (2003), Ellul, Shin, and Tonks (2005)). However, much of this literature is dated and is based on research conducted well before the rapid growth in the number of HFTs over the past two decades. More recent studies of the pre-opening period include Bellia, Pelizzon, Subrahmanyam, Uno, and Yuferova (2017) and Anagnostidis, Fontaine, and Varsakelis (2020) who analyze the role of HFTs as a group in price discovery during the pre-opening period; and Daures, Moinas, and Boussetta (2025) who investigate the role of the pre-opening period in fragmented markets, with one market featuring a pre-opening period while the other does not. Although this is not the main focus of our analysis, we also document that HFTs contribute to price discovery in the pre-opening period. We complement the existing literature, thanks to the granularity of our data, which allows us to directly trace individual traders, by investigating the type of HFTs that choose to participate in the pre-opening period, their potential motives to do so, and the association between this decision and market liquidity in the continuous trading period. In particular, we show that HFTs that choose to participate in the pre-opening period exploit the pre-opening order time priority, and their preferred trading strategy in the continuous period that follows is market-making rather than opportunistic trading.

2 Hypotheses

In this section, we develop the hypotheses that would allow us to answer the following questions: (i) which type of HFTs chooses to come to the pre-opening period: opportunistic or market makers; (ii) why a particular type of HFTs chooses to come to the pre-opening period; and (iii) whether the decision to come to pre-opening period is *associated* with improved or degraded market liquidity.

2.1 Trader type: market-making vs. opportunistic HFTs

We begin by mapping the heterogeneity of HFTs' strategies in the continuous trading period (SEC (2010) and Hagströmer and Norden (2013)) to their decision to come to the

pre-opening period. In particular, opportunistic HFTs are characterized by engagement in price discovery through their trades and the active use of market (marketable limit) orders for trading, while market-making HFTs are characterized by engagement in price discovery through quote adjustment to avoid being picked off by opportunistic trades, active use of liquidity-providing (i.e., passive) orders for trading, and continuous presence at the top and the best bid-offer level of the limit order book. We use these characterizations to test the following alternative hypotheses:

Hypothesis 1. *Opportunistic HFTs are more likely to come to the pre-opening period. In other words, Active HFTs are opportunistic traders.*

Hypothesis 2. *Market-making HFTs are more likely to come to the pre-opening period. In other words, Active HFTs are market-makers.*

2.2 Motives: information vs. queue position

[Medrano and Vives \(2001\)](#) show that strategic informed traders might have a reason to come to the pre-opening period, which is relevant for opportunistic HFTs. As a consequence, if HFT's participation is driven by the information channel, it is natural to expect that they will participate in the pre-opening more often when information intensity increases.

Hypothesis 3. *The likelihood of HFT participation in the pre-opening period increases on days characterized by high information intensity, such as following earnings announcements, periods of high volatility, large absolute overnight returns, or short-term return reversals.*

In addition, we also test the presence of the information channel indirectly. In case Active HFTs are indeed using the pre-opening period to gain an informational advantage over other market participants, this participation should be reflected in their profits.

Hypothesis 4. *Active HFTs earn larger profits in the continuous trading period than Non-Active HFTs.*

Furthermore, in the spirit of [Menkveld and Zoican \(2017\)](#) who model HFT’s type as a deterministic function of their queue position, we test whether securing a superior queue position can serve as a driving force behind the participation decision. We note that limit orders transferred from the pre-opening period to the continuous trading period will have higher time priority than orders entered during the continuous trading period. So, HFTs might use the pre-opening period to secure their place in the limit order queue (see [Li, Wang, and Ye \(2021\)](#), [Goldstein, Kwan, and Philip \(2023\)](#), [Dahlström, Hagströmer, and Nordin \(2024\)](#) for the importance of queue position for strategic order submission / cancellation decisions).

Hypothesis 5. *HFTs use the pre-opening period to secure time priority in the continuous trading period.*

While Hypotheses 3 and 4 are consistent with the information channel, Hypothesis 5 is consistent with the liquidity provision channel.

2.3 Participation and market liquidity

After mapping the heterogeneity of HFTs’ strategies to the decision to come to the pre-opening period and investigating the reasons behind it, we investigate whether such a decision is associated with improved or deteriorated market liquidity by testing the following hypotheses:

Hypothesis 6. *The HFT’s decision to come to the pre-opening period is associated with improved market liquidity.*

3 Institutional structure and data description

3.1 Institutional structure

The daily trading schedule of the TSE is characterized by two separate trading periods, one in the morning and one in the afternoon, along with two pre-opening periods (one early

in the morning and the other in the middle of the day).⁴ Figure 1 provides a representation of the different trading periods. In both the morning and afternoon periods, the opening price is determined by a single price auction (*Itayose* in Japanese) that kicks off at 9:00 a.m. for the morning session and at 12:30 p.m. for the afternoon session, based on buy and sell orders accumulated during the pre-opening periods that precede them.⁵

INSERT FIGURE 1 HERE

The opening time on the TSE is not randomized, as happens on some European exchanges such as the Deutsche Börse. As soon as the opening price of a stock is determined, the continuous trading period for that stock commences. Thus, these three segments, the pre-opening period, the opening call auction, and the continuous trading period, are seamlessly connected. In the pre-opening period, investors are free to submit orders as they do in the continuous trading period. There are two general types of orders allowed on the TSE: limit orders and market orders. Each trading day, the TSE starts receiving orders from its member-brokers at 8:00 a.m. (12:05 p.m. for the afternoon session) and does so without executing any orders until the single price auction for the market opening begins at 9:00 a.m. (12:30 p.m.).

As soon as the TSE receives orders, it disseminates the pre-opening quotes, not just the best ask and best bid, but 10 quotes above and below the best quotes, to the market.⁶ Every time the exchange receives an order, the pre-opening quotes are refreshed. The best ask (bid) is identified as the smallest (largest) ask (bid) price at which the cumulative depth of the ask (bid) schedule is greater than the cumulative depth of the bid (ask) schedule.⁷ The

⁴In Japan, there are two private trading venues, SBI Japannext and ChiX-Japan, whose trading periods start before 9:00 a.m.; however, very little trading occurs before 9:00 a.m. It should be mentioned that trading in the Nikkei Stock Index Futures on the Singapore Stock Exchange (SGX) starts at 8:45 a.m., Tokyo time; such trading may also contribute to price discovery.

⁵Single price auctions are also used to determine closing prices. The rules for the establishment of the closing call auction price are slightly different from those for the opening call auction price (see TSE (2015)).

⁶A subscriber to the “full quotes” service would be able to see price and quantity information on the entire order book. However, the prices and quantities for the best ask and the best bid are the same as for the standard service.

⁷A detailed description of the determination of the best bid and ask and the rules are provided in TSE (2015).

principle underlying order matching is based on price and time priority in the continuous trading period. In the opening call auction, however, time priority is ignored. That is, all orders placed in the pre-opening period, before the opening price has been determined, are regarded as simultaneous.

The opening auction determines the price at which the largest number of executions (by number of shares) is possible. There are three conditions to be met at the auction: (1) all market orders must be executed at the opening price; (2) orders with a sell limit price lower than the opening price and buy limit price higher than the opening price must be executed; (3) buy and sell orders with limit prices equal to the opening price must be executed for the entire amount on either the buy or the sell side. This third condition suggests that orders on either side, whose limit prices are equal to the opening price, are often not fully executable. If this happens, the TSE allocates the available shares to participating member firms on a pro-rated basis. As explained above, the speed (or timing) of order submission does not matter during the pre-opening period because of the lack of time priority in the opening call auction. Therefore, investors care about the timing of their submission only due to the treatment of unfilled orders in the continuous trading period, when time priority is activated.

3.2 Data

We use order data covering the complete history of an order (new order entry, execution, revision of quantity or price, and cancellation in the pre-opening period and continuous trading period) obtained from the TSE for April 2013 for the TOPIX100 index.⁸ This index is comprised of stocks from the TSE’s First Section with high liquidity and comprises over 60% of TSE market capitalization.⁹ Of the TOPIX100 stocks, we exclude three (Murata, Nintendo, and Nihon Densan) that have larger trading volumes on exchanges other than the

⁸Other papers using data from the TSE are [Goshima, Tobe, and Uno \(2019\)](#) and [Yamada \(2021\)](#).

⁹Stocks listed on the TSE are split into different sections based on their market capitalization, number of shareholders, and other parameters. The TSE’s First Section includes relatively large companies and comprises about 94.8% of the overall trading volume and 98.6% of the market capitalization.

TSE, since the focus of our study is the trading system on this exchange. We also exclude from our sample stock-days for which special quotes are disseminated before or during the single price auction. During those stock-days, orders submitted during the pre-opening period do not have to meet the normal opening price rules. Each historical record in our data is time-stamped at the millisecond level and includes information on order type, side (buy or sell), number of shares, limit price, unique order number, and virtual server (VS) ID.¹⁰

We find that 3,661 VSs were used in the sample period considered. However, we observe that orders pertaining to the same stock are sometimes submitted by different VSs; for example, an order submitted by VS “A” was canceled by VS “B.” By combining all the order and trade information from the market in our sample period, we identify 2,128 groups of VSs, which we call “traders” because they represent trading desks. Figure 2 depicts trader size based on the number of VSs they employ. Thanks to our matching procedure, our database is similar to one with anonymous trader identifiers, because we can actually track the traders’ behavior through time.¹¹

INSERT FIGURE 2 HERE

4 Identification of Active and Non-Active HFTs

As a first step, we classify traders as HFTs and Non-HFTs. A useful guideline defining the features of HFTs has been presented by the U.S. Securities and Exchange Commission (SEC). That document, (SEC, 2010, p. 45), lists five characteristics of HFTs:

1. *“Use of extraordinarily high speed and sophisticated programs for generating, routing, and executing orders.”*

¹⁰The VS is the logical path established between TSE’s system (the matching engine) and a user’s system to communicate with each other.

¹¹For studies in the HFT literature that use a database with user/trader identifier see, e.g., Hagströmer and Norden (2013), Menkveld (2013), Brogaard, Hagströmer, Norden, and Riordan (2015), Baron, Brogaard, Hagströmer, and Kirilenko (2019), and Korajczyk and Murphy (2019). The data typically come from Nordic (NASDAQ OMX) or Canadian (Toronto Stock Exchange) markets, except Menkveld (2013), which focuses on the Dutch market. However, all these papers focus only on the continuous trading period in their analysis.

2. *“Use of co-location services and individual data feeds offered by exchanges and others to minimize network and other latencies.”*
3. *“Very short time-frames for establishing and liquidating positions.”*
4. *“Submission of numerous orders that are canceled shortly after submission.”*
5. *“Ending the trading day in as close to a flat position as possible (that is, not carrying significant, unhedged positions overnight).”*

Motivated by this list of characteristics, we use both speed and inventory to classify traders: speed matches characteristics 1, 2, and 4 above, while inventory matches characteristics 3 and 5.

Our data permit us to use these criteria to classify traders as HFTs. They enable us to (i) track trader behavior, (ii) identify HFTs, (iii) cluster traders into several groups, such as HFTs and Non-HFTs, and (iv) classify them as Active and Non-Active traders based on their participation in the pre-opening period.

As a measure of speed, we empirically measure the minimum elapsed time between two consecutive order submissions for the same stock, without any restrictions, for a combination of two order types (i.e., any two out of new orders, cancellations, and revisions during our sample period).¹² We refer to our measure of time elapsed, using the time stamp of the order, as speed, so a high speed means a short time between order submissions. Some authors refer to this concept as latency, but we prefer to use speed since we view latency as having connotations of the technological limits of speed, rather than the actual speed realized. Henceforth, we simply call our measure “speed.” A realization of a very high speed has to be supported by the appropriate trader’s infrastructure. Hence, the number of VSs a trader uses is a crucial determinant of her speed.

The other major classification variable we employ is the trader’s inventory, which is estimated as the (absolute) ratio of the buy volume minus the sell volume at the end of day

¹²Our measure of speed is similar in spirit to the “strategic runs” (linked submissions, cancellations, and executions that are likely to be part of a dynamic strategy) used by [Hasbrouck and Saar \(2013\)](#). However, their data do not enable them to identify strategic runs coming from individual traders.

k , divided by the total trading volume of that trader on that day. Many empirical studies report that the key characteristic of HFT liquidity providers is a flat inventory position at the end of each trading day (Menkveld (2013), Kirilenko, Kyle, Samadi, and Tuzum (2017), and SEC (2014)). To investigate this issue further, we compute the end-of-day inventory for each trader and each stock, as well as the intraday inventory, the volatility of intraday inventory, and the number of times during the day that the inventory crosses zero, per trader-stock-day.

We classify a trader as an HFT if and only if that trader satisfies the following criteria:

1. A trader should be present on the market in at least 25% of all eligible stock-days.¹³
2. For at least 75% of the stock-days a trader chooses to be present on the market,
 - (a) the trader should exhibit speed below the median (excluding traders with speed greater or equal to 60 seconds since those are likely to be humans).¹⁴
 - (b) the trader should have an inventory that is less than the median (excluding traders with inventory equal to 100% since those are likely to be buy-and-hold investors and traders with zero trading volume).

Overall, this classification scheme results in 13 traders classified as HFTs in our sample.¹⁵ In addition, we divide traders into Active and Non-Active based on their decision to participate

¹³We consider a trader to be present in the market if a trader has executed at least one transaction during the day or submitted at least two messages during the continuous trading period.

¹⁴Our definition of speed is not nominal speed but relative speed. This is in line with what Baron, Brogaard, Hagströmer, and Kirilenko (2019) find: that it is relative speed, and not nominal speed, that drives differences in performance across HFTs. It is not being fast in absolute terms that allows an HFT to capture trading opportunities but being fastest in relative terms. This is also in line with the theoretical literature on HFTs; see Biais, Foucault, and Moinas (2015), Budish, Cramton, and Shim (2015), and Menkveld and Zoican (2017).

¹⁵We have performed classification sensitivity analysis by tweaking one of the three thresholds at a time (i) % of eligible stock-days a trader should be present on the market (baseline of 25%), (ii) % of stock-days a trader chooses to be present on the market with speed and inventory below threshold (baseline of 75%), (iii) speed and inventory percentile threshold (baseline 50%). By increasing the first threshold and requiring a trader to be present on the market for 35% of eligible stock-days we identify exactly the same 13 trading desks as HFTs. By increasing the second threshold and requiring a trader to exhibit HFT characteristics in 80% of the stock-days a trader chooses to be on the market we identify 11 trading desks as HFTs (a subset of the original 13 desks). By decreasing the speed and inventory threshold to 40% we identify the same 11 trading desks as HFTs. These two desks are always classified as Non-Active HFTs, and do not contribute to the price discovery and liquidity provision in the morning continuous trading period.

in the pre-opening period. We note that while traders are permanently classified as HFTs or Non-HFTs, and do not switch their type throughout the sample, their classification into Active and Non-Active traders can vary from one stock-day to another.

We refer to Figure 3 for the average proportion of traders in each of the trader categories. We note that, on average, only 1.5% of traders per stock-day are classified as HFTs, which implies that only around 20% of HFTs ($\frac{0.3\%}{0.3\%+1.2\%}$) choose to participate during the morning pre-opening period.

INSERT FIGURE 3 HERE

Table 1 summarizes the traders' characteristics in terms of speed (Panel A) and inventory (Panel B), based on the following procedure. First, we compute the speed and inventory measures for each trader-stock-day. Second, we compute median speed and inventory measures for each trader category, stock, and date. Finally, we compute the median across stock-days for each trader category. We note that the median numbers of Active and Non-Active HFTs per stock-day are two and six traders, respectively. We show that the median minimum speed is 0.001 seconds for both Active and Non-Active HFTs alike.¹⁶ The corresponding median minimum speeds for Active and Non-Active Non-HFTs are 4.993 and 1.018 seconds, respectively. Median end-of-day inventories for Active and Non-Active HFTs (Non-HFTs) are 1.6% and 5.1% (57.3% and 75.0%), respectively. Overall, we observe that HFTs are much faster and hold less inventory than Non-HFTs.

We also note that speed characteristics are quite similar between Active and Non-Active HFTs, suggesting that both trader groups have access to similar technology and infrastructure. At the same time, Active HFTs tend to have lower inventory than Non-Active HFTs, potentially indicating the differences in the trading strategies employed by the two groups. This feature is also highlighted by the fact that the inventory of Active HFTs tends to cross zero five times more often than that of Non-Active HFTs. This finding indicates that Active HFTs tilt toward market-making strategies.

¹⁶Our data are time-stamped at the millisecond level, so the minimum speed cannot be estimated below 0.001 seconds.

INSERT TABLE 1 HERE

5 Empirical results

Using desk-level data that no prior paper has had across the pre-opening and continuous trading periods, we test the six hypotheses reported in Section 2 to disentangle the following questions (i) which type of HFTs chooses to come to the pre-opening period (Section 5.1), (ii) why some type of HFTs chose to come to pre-opening period (Section 5.2); and (iii) whether there is an *association* between improved market liquidity and HFT’s decision to come to the pre-opening period (Section 5.3).

5.1 Which type of HFTs chooses to come to the pre-opening period?

The first set of tests allows us to map the heterogeneity of HFTs’ strategies in the continuous trading period to their decision to come to the pre-opening period. We analyze the contribution of Active and Non-Active HFTs to price discovery (Section 5.1.1), the use of liquidity-providing orders for trading (see Section 5.1.2), and the presence at the various levels of the limit order book (see Section 5.1.3).

5.1.1 Contribution to price discovery

We begin by analyzing the price discovery contribution of Active and Non-Active HFTs during the continuous trading period. Opportunistic traders are likely to have a larger price impact via their transactions, while market-makers can contribute to price discovery via quote-adjustments to decrease the pick-off risk.

It is reasonable to assume that both the quoted and traded prices of a stock will change in response to the arrival of new information. However, while the traded price is subject to bid-ask bounce issues, the midquote is not. Therefore, we use the midquote as a proxy for a stock’s fundamental value. There are two types of information events, defined as messages, that could move the midquote: (i) messages that do not result in an actual transaction but

do have the potential to affect the midquote (referred to below as aggressive messages)¹⁷ and (ii) messages that result in the transaction (referred to below as trades).¹⁸

For the purpose of further analysis, we focus solely on the morning pre-opening period and the morning continuous trading period because they provide the earliest opportunity for market prices to reflect information accumulated after the prior day’s closing until the morning of the trading day.¹⁹ Traders are likely to reflect this information in their bids and offers in the morning pre-opening period and in the early part of the continuous trading period. In addition, we exclude the first 10 minutes of the morning pre-opening period because during that period, the limit order book accumulates many orders that were waiting overnight for the beginning of the morning pre-opening period. The arrival times of these orders are not directly related to the traders’ actual submission decisions.

In order to reliably estimate contributions to price discovery by various trader types, we also require that there be at least five aggressive messages from each of the trader categories during the morning pre-opening period and at least five aggressive messages and five trades from each of the trader categories during the morning continuous trading period for a stock-day to be included in the analysis. This filter leaves us with 19 trading days and 79 stocks in the sample. The number of eligible stocks per trading day varies considerably, ranging from 13 to 47 stocks per day (with a mean of around 30 stocks). Some stocks enter the sample for only a single trading day, while others are present for 18 trading days in the sample. Overall, our sample consists of 578 stock-days.²⁰

¹⁷During the morning pre-opening period, such messages are as follows: (i) the submission, cancellation, or quantity modification of any market order; (ii) the submission, cancellation, price modification, or quantity modification of a limit buy (sell) order with a limit price greater (less) than or equal to the prevailing best bid (ask). During the morning continuous trading period, such messages are the submission, cancellation, or price modification of a limit buy (sell) order with a limit price greater (less) than or equal to the prevailing best bid (ask) but not initiating the transaction.

¹⁸Since there is no trading during the morning pre-opening period, there are no messages that result in transactions to analyze. During the morning continuous trading period, such messages are market and marketable limit orders initiating transactions.

¹⁹Since the Japanese market opens when the major markets in Europe, North America, and elsewhere in Asia, have been closed, some for several hours, this information gap is particularly long.

²⁰A sufficient presence of Active HFTs is the main reason for the decrease in the number of eligible stock-days.

We follow [Hasbrouck \(1991\)](#) and [Brogaard, Hendershott, and Riordan \(2019\)](#) and estimate the SVAR model to quantify the informational content of aggressive messages and trades. SVAR allows for both lagged and contemporaneous effects of aggressive messages and trades on returns. SVAR models are estimated in event time for each stock-day and period, where the event can be any message submitted:

$$\begin{aligned}
r_t &= \sum_{l=1}^5 \alpha_l^1 r_{t-1} + \sum_{l=0}^5 \beta_l^{1,1} X_{t-1}^1 + \sum_{l=0}^5 \beta_l^{1,2} X_{t-1}^2 + \dots + \sum_{l=0}^5 \beta_l^{1,8} X_{t-1}^8 + \epsilon_t^1 \\
X_t^1 &= \sum_{l=1}^5 \alpha_l^2 r_{t-1} + \sum_{l=1}^5 \beta_l^{2,1} X_{t-1}^1 + \sum_{l=1}^5 \beta_l^{2,2} X_{t-1}^2 + \dots + \sum_{l=1}^5 \beta_l^{2,8} X_{t-1}^8 + \epsilon_t^2 \\
X_t^2 &= \sum_{l=1}^5 \alpha_l^3 r_{t-1} + \sum_{l=1}^5 \beta_l^{3,1} X_{t-1}^1 + \sum_{l=1}^5 \beta_l^{3,2} X_{t-1}^2 + \dots + \sum_{l=1}^5 \beta_l^{3,8} X_{t-1}^8 + \epsilon_t^3 \\
&\dots \\
X_t^8 &= \sum_{l=1}^5 \alpha_l^9 r_{t-1} + \sum_{l=1}^5 \beta_l^{9,1} X_{t-1}^1 + \sum_{l=1}^5 \beta_l^{9,2} X_{t-1}^2 + \dots + \sum_{l=1}^5 \beta_l^{9,8} X_{t-1}^8 + \epsilon_t^9
\end{aligned} \tag{1}$$

where r_t is the log-return based on the midquote. The variables $X_t^1 - X_t^8$ are defined as aggressive messages from Active HFTs (X_t^1), trades initiated by Active HFTs (X_t^2), aggressive messages from Non-Active HFTs (X_t^3), trades initiated by Non-Active HFTs (X_t^4), aggressive messages from Active Non-HFTs (X_t^5), trades initiated by Active Non-HFTs (X_t^6), aggressive messages from Non-Active Non-HFTs (X_t^7), and trades initiated by Non-Active Non-HFTs (X_t^8). All these variables are signed directional activity measures taking the value of +1 for messages that have the potential to push the midquote up and -1 for messages that have the potential to push the midquote down.²¹

For each stock, day, and trading period, we compute cumulative structural impulse response functions (SIRFs) for returns for 20 steps ahead for +1 unit shock to directional activity measures and winsorize them at the 1% and 99% levels. We report the average of the respective cumulative SIRFs for Active and Non-Active HFTs only in Panel A of Table 2. We note that the SVAR model results can be interpreted as an effect on the permanent price component without explicitly modeling the dynamics of the transitory price component as in a state-space model representation and are much easier to estimate than the latter. We

²¹We note that SVAR models estimated for the pre-opening period include only a subset of the $X_t^1 - X_t^8$ variables.

refer to the additional analysis performed in Section 6.1 for the state-space approach.

INSERT TABLE 2 HERE

A +1 unit shock to aggressive messages submitted by Active HFTs during the morning pre-opening period leads to a price impact of 0.233 bps. During the morning continuous trading period, trades of Non-Active HFTs have the largest price impact of 6.261 bps, which is 1.4 times larger than the price impact of trades initiated by Active HFTs (4.434 bps).²² By contrast, the aggressive messages (i.e., quotes) of Active HFTs have a larger price impact (1.282 bps) than the aggressive messages of Non-Active HFTs (0.555 bps).

Table 2 Panel B compares structural impulse response functions from Eq. (1) for Active and Non-Active HFTs in the following regression framework:

$$\begin{aligned}
SIRF_{c,s,d}^{imp} = & \eta_s + \xi_d + \\
& + Active_{c,s,d} + \\
& + \beta_1 Return\ Reversal_{s,d} + \beta_2 |Overnight\ Return_{s,d-1 \rightarrow d}| + \\
& + \beta_3 Earnings\ Announcement_{s,d-1 \rightarrow d} + \\
& + \beta_4 Range_{s,d-1} + \beta_5 Closing\ Quoted\ Spread_{s,d-1} + \\
& + \epsilon_{c,s,d}
\end{aligned} \tag{2}$$

where $SIRF_{c,s,d}^{imp}$ is the SIRF computed for returns and cumulated over 20 steps ahead for +1 shock to directional activity (*imp*). Directional activity measures include aggressive messages from Active and Non-Active HFTs and trades initiated by Active and Non-Active HFTs. We regress SIRFs on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return,

²²For comparison, Brogaard, Hendershott, and Riordan (2019) report that HFT trades that change the midpoint have a price impact of 3.25 bps using a similar framework during the continuous trading period for Canadian stocks.

a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. We also include stock and day fixed effects and cluster standard errors by stock.

We show that aggressive messages of Active HFTs are systematically more informative than those of Non-Active HFTs by 0.727 bps. This shows that information gathered by Active HFTs during the pre-opening period is used to adjust their quotes to incorporate relevant information. On the contrary, the picture is very different for trades. The trades of Active HFTs are less informative than those of Non-Active HFTs by 1.826 bps. We note that the difference in SIRFs between Active and Non-Active HFTs is very stable across various specifications: without fixed effects, with stock and day fixed effects, and with fixed effects and controls. Overall, the price impact of trades (reaching 6.261 bps for Non-Active HFTs) is larger than those of aggressive messages (reaching 1.282 bps for Active HFTs), which undermines the idea of Active HFTs having a larger contribution to the price discovery process in the morning continuous trading period. To sum up, price discovery analysis points in favor of Hypothesis 2 and rejects Hypothesis 1.

5.1.2 Usage of liquidity-providing orders for trading

Next, we check whether Active HFTs mainly use liquidity-providing orders for trading. We compute the volume imbalance ratio, measured as a difference between trading volume when HFTs provide (i.e., do not initiate the transaction) and demand liquidity (i.e., do initiate the transaction) from the market:

$$Imb_{k,s,d} = \frac{Trading\ Volume_{k,s,d}^{Supply} - Trading\ Volume_{k,s,d}^{Demand}}{Trading\ Volume_{k,s,d}^{Supply} + Trading\ Volume_{k,s,d}^{Demand}} \quad (3)$$

where $Trading\ Volume_{k,s,d}^{Supply}$ ($Trading\ Volume_{k,s,d}^{Demand}$) is the trading volume measured either in JPY or shares when trader k provides (demands) liquidity for stock s and day d . Trader k demands liquidity if trader k initiates the transaction (i.e., uses a market or marketable limit order) and provides liquidity if she does not initiate the transaction. Afterwards,

we compute the average volume imbalance ratio for each trader category (Active HFTs vs. Non-Active HFTs), stock, and date.

Table 3 Panel A provides summary statistics for the imbalance ratio measured based on the JPY trading volume and share trading volume for Active and Non-Active HFTs. We show that Non-Active HFTs are neutral with respect to liquidity provision, i.e., they demand the same amount of liquidity as they provide, while Active HFTs have a trading volume imbalance ratio around 67% indicating that they provide liquidity for the majority of their transactions.

INSERT TABLE 3 HERE

Table 3 Panel B compares trading imbalance ratio of Active and Non-Active HFTs in the following regression framework:

$$\begin{aligned}
Imb_{c,s,d} = & \eta_s + \xi_d + \\
& + Active_{c,s,d} + \\
& + \beta_1 Return\ Reversal_{s,d} + \beta_2 |Overnight\ Return_{s,d-1 \rightarrow d}| + \\
& + \beta_3 Earnings\ Announcement_{s,d-1 \rightarrow d} + \\
& + \beta_4 Range_{s,d-1} + \beta_5 Closing\ Quoted\ Spread_{s,d-1} + \\
& + \epsilon_{c,s,d}
\end{aligned} \tag{4}$$

where $Imb_{c,s,d}$ is the volume imbalance ratio of trader category c for stock s on day d . We regress it on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. We also include stock and day fixed effects and cluster standard errors by stock.

We note that Active HFTs have a consistently larger imbalance ratio based on JPY trading volume (share trading volume) by 70.001% (69.999%) than Non-Active HFTs independently of the specification. To sum up, we show that our conclusions regarding the usage of liquidity-providing orders for trading support Hypothesis 2 and reject Hypothesis 1.

5.1.3 Passive presence in the limit order book

To further test Hypothesis 2, we check whether Active HFTs are present more often in the limit order book than Non-Active HFTs. Following Hagströmer and Norden (2013), we measure presence at the various levels of the limit order book during the morning continuous trading period. First, we rebuild the limit order book for the morning continuous trading period at the millisecond level. Second, we use a time-weighted average for each trader to compute how often that trader is at the top of the limit order book (as the trader with the highest time priority at the best bid-offer level), at the best bid-offer level, and at the best three levels of either side of the limit order book.

INSERT TABLE 4 HERE

Table 4 Panel A provides summary statistics for passive presence of HFTs at the top, best bid-offer level, and three best levels of the order book. Active HFTs are present more often at the top and the best bid-offer level of the limit order book than are Non-Active HFTs. More specifically, on average, each trader from the Active HFT (Non-Active HFT) category is present 8.263% (5.186%) at the top of the limit order book and 34.457% (28.478%) at the best bid-offer level. Those Non-Active HFTs catch up with Active HFTs in terms of presence only at the deeper levels of the order book, as can be seen from their statistically indistinguishable presence at the three best levels of the order book of 57.013% and 56.460%, respectively.

Table 4 Panel B compares contribution to the limit order book for Active and Non-Active HFTs in the following regression framework:

$$\begin{aligned}
Presence_{c,s,d}^{level} = & \eta_s + \xi_d + \\
& + Active_{c,s,d} + \\
& + \beta_1 Return\ Reversal_{s,d} + \beta_2 |Overnight\ Return_{s,d-1 \rightarrow d}| + \\
& + \beta_3 Earnings\ Announcement_{s,d-1 \rightarrow d} + \\
& + \beta_4 Range_{s,d-1} + \beta_5 Closing\ Quoted\ Spread_{s,d-1} + \\
& + \epsilon_{c,s,d}
\end{aligned} \tag{5}$$

where $Presence_{c,s,d}^{level}$ is passive presence of trader category c for stock s on day d measured at different levels of the limit order book (*level*). We regress it on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. We also include stock and day fixed effects and cluster standard errors by stock.

Active HFTs are more present at the top and best bid-offer level by 3.077% and by 5.979%, respectively, than Non-Active HFTs independently of the specification. In other words, Active HFTs are present at the top and best bid-offer level of the limit order book 1.6 and 1.2 times more often, respectively, than Non-Active HFTs. To sum up, this analysis also points in favor of failing to reject Hypothesis 2.

Overall, we map heterogeneity of HFT's strategies to their decision to participate in the pre-opening period and conclude that Active HFTs are largely market-makers, while Non-Active HFTs are opportunistic traders.

5.2 Why do market-making HFTs choose to come to the pre-opening period?

The second set of tests is related to the reasons of market-making HFTs to come to the pre-opening period. Figure 4 shows that of all orders submitted during the pre-opening period by HFTs, 65.8% are canceled before the opening call auction. Among the orders that do make it to the opening auction, 9.8% are executed during the opening auction. Around 10.3% of orders would be executed during the continuous trading period when time priority is activated, thus allowing market makers to secure their place in the limit order queue. The remainder of the orders are canceled by the trader or the TSE in the continuous period that follows.

We zoom into pre-opening period in order to understand better the dominant behavior of HFTs, namely, a high cancellation rate. In particular, Figure 5 shows the submission patterns of new orders and cancellations throughout the pre-opening period. Almost all orders stemming from HFTs that are entered before 8:59 are canceled (cancellation rate reaches almost 100%). The high cancellation rate of the early orders is not surprising, since HFTs react to the order flow observed during the pre-opening period and re-adjust their quotes. In the last minute before the opening auction the cancellation rate drops to around 40%, and resembles the cancellation rate observed during the morning continuous period of around 35% and therefore is not unusual. This behavior could be consistent with both the information channel and the queue position securing channel. It is consistent with the information channel: by posting orders they do not plan to execute, HFTs try to provoke other traders to reveal their intentions. It is also consistent with the queue position securing: a market-maker who desires time priority in the continuous period but does not want auction execution can cancel and re-submit the order just outside the likely clearing range.

INSERT FIGURES 4 – 5 HERE

In the following analysis we aim to disentangle among the two channels. In particular,

we analyze whether this decision is consistent with (i) the information channel, namely, heterogeneity in information intensity and profits made (see Section 5.2.1) and/or (ii) the liquidity provision channel, namely, securing a superior position in the limit order queue (see Section 5.2.2).

5.2.1 Information channel

We begin by relating the HFT's decision to participate in the pre-opening period to the intensity of information environment. If HFTs indeed use the pre-opening period to gather information, it is natural to assume that they would be more likely to do so when overnight information intensity is large. We proxy for the variation in the overnight information environment by absolute overnight returns, the return reversal, overnight earnings announcements (34 announcements in total), and volatility. In particular, we estimate the following linear probability model for individual HFTs (13 trading desks) for the stock-days they were present on the market:

$$\begin{aligned}
y_{k,s,d} = & \alpha_k + \eta_s + \xi_d + \\
& + \beta_1 \textit{Return Reversal}_{s,d} + \beta_2 |\textit{Overnight Return}|_{s,d-1 \rightarrow d} + \\
& + \beta_3 \textit{Earnings Announcement}_{s,d-1 \rightarrow d} + \\
& + \beta_4 \textit{Range}_{s,d-1} + \beta_5 \textit{Closing Quoted Spread}_{s,d-1} + \\
& + \beta_6 |\textit{Cumulative EOD Inventory}|_{k,s,d-1} + \epsilon_{k,s,d}
\end{aligned} \tag{6}$$

where $y_{k,s,d}$ is either a dummy variable that takes value of one, if HFT k is active in the pre-opening period of stock s on day d , and zero otherwise; or a dummy variable that takes value of one, if HFT k switches from Non-Active HFT for stock s and day $d - 1$ to Active HFT for stock s on day d , and zero otherwise; or % of messages coming from HFTs during the pre-opening period. Independent variables are a dummy variable that takes value of one, if overnight return and return over first 30 minutes are of the opposite signs for stock s on day d ($\textit{Return Reversal}_{s,d}$), the absolute value of overnight return for stock s from market

close on day $d - 1$ to market open on day d ($|Overnight\ Return|_{s,d-1 \rightarrow d}$), a dummy variable that takes value of one, if there was an earnings announcement for stock s from market close on day $d - 1$ to market open on day d ($Earnings\ Announcement_{s,d-1 \rightarrow d}$), volatility of stock s on day $d - 1$ proxied by high minus low range ($Range_{s,d-1}$), liquidity of stock s on day $d - 1$ proxied by closing bid-ask spread ($Closing\ Quoted\ Spread_{s,d-1}$), and absolute value of cumulative end-of-day inventory (in 100 mln JPY) for HFT k , stock s , and day $d - 1$ ($|Cumulative\ EOD\ Inventory|_{k,s,d-1}$).²³ Depending on the specification, the regression includes either trader fixed effects (α_k), stock fixed effects (η_s), day fixed effects (ξ_d), or all of them. Standard errors are clustered by stock.

INSERT TABLE 5 HERE

Table 5 shows the result of estimating Eq. (6). Among variables related to various information intensity (return reversal, absolute overnight return, overnight earnings announcement, and volatility), only absolute overnight return consistently shows up significantly. However, the sign is counterintuitive. Focusing on the strictest specification that includes trader, stock, and day fixed effects, a one standard deviation increase in absolute overnight return (0.64%) is associated with a decrease in the probability of being an Active HFT (of switching from Non-Active to Active HFT) by 1.15% (1.02%).²⁴ The % of messages in the pre-opening period coming from HFTs is also negatively related to the size of overnight return. Overall, we do not find support for Hypothesis 3 that the likelihood of individual HFT's participation in the pre-opening period increases with information intensity.

We have excluded obvious reasons related to differences in the information environment, but we could not exclude that HFTs might use the pre-opening period to gather information. If this is indeed true, it should be reflected in the profits they make. Table 6 compares profits

²³We control for the previous day's inventory since HFTs might use the opening auction to offload the undesired inventory position from the previous day; however, we also note that HFTs end up their day flat, so this is an unlikely reason to drive the participation decision.

²⁴Unconditional probability of being an Active HFT (to switch from Non-Active to Active HFT) is 19.43% (19.41%).

as of 09:15 am, 09:30 am, and 09:45 am, for Active and Non-Active HFTs. First, we compute relative profits for each trader-stock-day:

$$Profit_{k,s,d}^{ref} = \frac{Sell\ Volume_{k,s,d}^{ref} - Buy\ Volume_{k,s,d}^{ref} + Position_{k,s,d}^{ref}}{Sell\ Volume_{k,s,d}^{ref} + Buy\ Volume_{k,s,d}^{ref}} \quad (7)$$

where $Sell\ Volume_{k,s,d}^{ref}$ ($Buy\ Volume_{k,s,d}^{ref}$) is the sell (buy) volume of trader k for stock s and day d from 09:00 till ref (in JPY), $Position_{k,s,d}^{ref}$ is the position of trader k for stock s and day d at ref evaluated at the midquote prevailing at ref (in JPY), and ref is the reference time (09:15, 09:30, or 09:45). Profits are computed under the assumption that a trader starts every stock-day with a zero inventory position. Second, we compute the average and relative profits for each trader category (Active HFTs vs. Non-Active HFTs), stock, and date.

Then, we estimate the following regression:

$$\begin{aligned} Profit_{c,s,d}^{ref} = & \eta_s + \xi_d + \\ & + Active_{c,s,d} + \\ & + \beta_1 Return\ Reversal_{s,d} + \beta_2 |Overnight\ Return|_{s,d-1 \rightarrow d} + \\ & + \beta_3 Earnings\ Announcement_{s,d-1 \rightarrow d} + \\ & + \beta_4 Range_{s,d-1} + \beta_5 Closing\ Quoted\ Spread_{s,d-1} + \\ & + \epsilon_{c,s,d} \end{aligned} \quad (8)$$

where $Profit_{c,s,d}^{ref}$ is the profit of trader category c for stock s on day d measured at reference time ref . We regress profits on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. We also include stock and day fixed effects and cluster standard errors by

stock.

We do not find any statistically significant difference in profits made by Active and Non-Active HFTs and thus, reject Hypothesis 4. We note that this result is consistent with the fact that Active HFTs do not have an informational advantage over Non-Active HFTs, but it is also consistent with an equilibrium outcome in which none of the HFT types dominate the other one in terms of performance.

Overall, we rule out the information-related channel as a reason behind the HFT's decision to participate in the pre-opening period.

5.2.2 Liquidity provision channel

In this section, we investigate whether the liquidity provision channel, namely, securing a superior queue position might be behind HFT's decision to participate in the pre-opening period. We link the proportion of orders that are entered during pre-opening period by Active HFTs, but executed in the continuous trading period to the passive presence of Active HFTs at various levels of the limit order book:

$$\begin{aligned}
Presence_{ActiveHFT,s,d}^{level} = & \eta_s + \xi_d + \\
& + EarlyOrder_{ActiveHFT,s,d} + \\
& + \beta_1 Return\ Reversal_{s,d} + \beta_2 |Overnight\ Return_{s,d-1 \rightarrow d}| + \\
& + \beta_3 Earnings\ Announcement_{s,d-1 \rightarrow d} + \\
& + \beta_4 Range_{s,d-1} + \beta_5 Closing\ Quoted\ Spread_{s,d-1} + \\
& + \epsilon_{ActiveHFT,s,d}
\end{aligned} \tag{9}$$

where $Presence_{ActiveHFT,s,d}^{level}$ is passive presence of Active HFTs for stock s on day d measured at different levels of the limit order book (*level*). We regress it on the percentage of early orders from Active HFTs that were executed during the continuous trading period for stock s and day d ($EarlyOrder_{ActiveHFT,s,d}$). Control variables include a dummy variable for return

reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. We also include stock and day fixed effects and cluster standard errors by stock.

INSERT TABLE 7 HERE

Table 7 shows that a one standard deviation increase in the proportion of such orders (11.238%) is associated with an increase of 3.034% (0.270×11.238) in time presence at the best bid-offer level, which is equivalent to 9% of the sample mean.²⁵ Thus, our findings are consistent with securing time priority and we do not reject Hypothesis 5. However, securing time priority does not fully explain the extent of liquidity provision by Active HFTs. We also note that our sample does not include any exogenous shock to the limit order priority rules, so this result should be interpreted with caution as we cannot claim causality.

5.3 HFT’s decision to come to the pre-opening period and market liquidity

In this section, we investigate whether Active HFTs are *associated* with improved market liquidity during normal and turbulent times alike. In particular, we test whether Active HFTs are associated with an improved quoted spread, maximum quoted depth between the best bid and offer levels of the limit order book, and depth imbalance between the best bid and offer levels of the limit order book (all measures are computed right before the transaction takes place and aggregated to a stock-date level using share-weighted average). We compute these measures using all the data and also the data without limit orders stemming from Active HFTs (under the assumption that other market participants continue to act in the same manner). We then run the following regression:

²⁵For the three best levels, a one standard deviation increase in the proportion of such orders is associated with an increase of 3.731% in time presence, which is equivalent to 7% of the sample mean.

$$\begin{aligned}
Liquidity_{c,s,d} = & \eta_s + \xi_d + \beta_1 Active_{c,s,d} + \\
& + \beta_2 Return\ Reversal_{s,d} + \beta_3 |Overnight\ Return|_{s,d-1 \rightarrow d} + \\
& + \beta_4 Earnings\ Announcement_{s,d-1 \rightarrow d} + \\
& + \beta_5 Range_{s,d-1} + \epsilon_{s,d}
\end{aligned} \tag{10}$$

where $Liquidity_{c,s,d}$ is either the quoted spread, quoted depth, or depth imbalance for stock s on day d , measured with or without Active HFT (c). We regress it on a dummy variable that takes value of one, if limit orders from Active HFTs are included in the data, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, and the previous day stock volatility proxied by high minus low range. We also include stock and day fixed effects and cluster standard errors by stock.

INSERT TABLE 8 HERE

Table 8 shows that the inclusion of Active HFTs is systematically associated with a decrease in the share-weighted quoted spread of 0.222 bps, an increase in the share-weighted quoted depth of 0.650 (in mln JPY), and a decrease in the share-weighted depth imbalance at the best bid-offer level of the limit order book of 0.076%. Thus, inclusion of Active HFTs is associated with improved market liquidity during normal times and hence, we cannot reject Hypothesis 6.

We also analyze whether differences in the market liquidity measures with and without Active HFTs can be driven by the securing queue position channel:

$$\begin{aligned}
\Delta Liquidity_{s,d}^{With-Without} = & \eta_s + \xi_d + \\
& + EarlyOrder_{ActiveHFT,s,d} + \\
& + \beta_1 Return\ Reversal_{s,d} + \beta_2 |Overnight\ Return|_{s,d-1 \rightarrow d} + \quad (11) \\
& + \beta_3 Earnings\ Announcement_{s,d-1 \rightarrow d} + \beta_4 Range_{s,d-1} + \\
& + \epsilon_{ActiveHFT,s,d}
\end{aligned}$$

where $\Delta Liquidity_{s,d}^{With-Without}$ is a difference in either the quoted spread, depth, or depth imbalance including and excluding limit orders from Active HFTs for stock s on day d . We regress it on the percentage of early orders from Active HFTs that were executed during the continuous trading period for stock s and day d ($EarlyOrder_{ActiveHFT,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, and the previous day stock volatility proxied by high minus low range. We also include stock and day fixed effects and cluster standard errors by stock.

INSERT TABLE 9 HERE

Table 9 shows that the securing queue position channel is associated with improved quoted depth and decreased depth imbalance in the limit order book. Even though the effect on the quoted spread is not statistically significant, the sign of the coefficient points in a direction consistent with this channel.

So far, we have checked whether the presence of Active HFTs is positively associated with market liquidity during normal times. Next, we analyze the behavior of Active HFTs during turbulent times. Following Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov (2018), we compute extreme price movements (EPMs) as a 99.9th percentile of absolute 10-sec midquote returns, and investigate how Active and Non-Active HFTs trade around EPMs. Supply (demand) is defined as the difference in the number of shares supplied (demanded)

in the direction vs. the opposite direction of EPM. Net share imbalance is a sum of supply and demand.²⁶ Liquidity provision is, therefore, defined as activity in the opposite direction of the EPM. Table 10 and Figure 6 show the results of the EPM analysis.

INSERT TABLE 10 and FIGURE 6 HERE

Within the 10-second interval when an EPM occurs, the net share imbalance of Active HFTs is against the direction of the EPM (-1,120 shares), while the opposite holds for Non-Active HFTs (+14,977 shares). Moreover, this effect is driven by liquidity-supplying orders for Active HFTs and liquidity-demanding orders for Non-Active HFTs.²⁷

In sum, the decision of HFTs to participate in the pre-opening period is positively associated with market liquidity both during normal and turbulent time periods and thus, we fail to reject Hypothesis 6.

6 Additional analysis

In this section, we provide several additional analyses. First, we explicitly model the permanent and transitory price components of stock price movements to strengthen our conclusions on the role of Active HFTs in price discovery (see Section 6.1). Second, we conduct subsample analysis (see Section 6.2) to ensure that our findings are robust.

6.1 State-space approach to price discovery

The SVAR approach used in the main analysis of the paper does not allow us to explicitly model both permanent and transitory components of price dynamics. We follow Brogaard, Hendershott, and Riordan (2014) and Menkveld and Saru (2024) and estimate the following

²⁶We have also repeated this analysis using relative share volume (number of shares relative to number of shares outstanding) and find that the results are qualitatively similar.

²⁷This analysis also speaks to the association between Active HFTs and volatility, since it shows that Active HFTs do not exacerbate extreme price movements. In the unreported analysis, we look at one-minute realized volatility with and without Active HFTs and do not find any significant difference between the two cases.

state-space model for price dynamics for each stock-day:

$$\begin{aligned}
p_t &= m_t + s_t \\
m_t &= m_{t-1} + \sum_{j \in J} \gamma_j \tilde{X}_t^j + \omega_t \\
s_t &= \phi s_{t-1} + \sum_{j \in J} \delta_j X_t^j + \epsilon_t
\end{aligned} \tag{12}$$

where p_t is the log-midquote, s_t is the pricing error (transitory component), m_t is the efficient price (permanent component), X_t^j are aggressive messages from Active HFTs, trades initiated by Active HFTs, aggressive messages from Non-Active HFTs, and trades initiated by Non-Active HFTs, respectively, and $\omega_t \sim N(0, \sigma_\omega^2)$ and $\epsilon_t \sim N(0, \sigma_\epsilon^2)$ are innovations in the efficient price (permanent component) and the pricing error (transitory component), respectively. All the X_t^j variables are signed directional activity measures taking the value of +1 for messages / trades that have the potential to push the midquote up, and -1 for messages / trades that have the potential to push the midquote down. Finally, $\tilde{X}_t^j$ are innovations in X_t^j from the vector-autoregression model with a lag of 10.

State-space models are estimated for the morning continuous trading period for each stock-day at the event time using the Kalman filter, where the event is any message with the potential to move the prevailing midquote. To estimate the model, we require at least 5 events associated with a change in the midquote for each of the X_t^j variables.²⁸ Due to the complexity of the state-space model estimation, we focus solely on the morning continuous trading period, since the morning pre-opening period lacks sufficient activity to obtain sensible estimates, and only on activity stemming from HFTs. Coefficient estimates for the efficient price and pricing error equations (γ and δ), winsorized at the 1% and 99% levels and averaged across stock-days are reported in Panels A and C of Table 11, respectively.

INSERT TABLE 11 HERE

We show that both the aggressive messages and the trades of Active and Non-Active

²⁸The sample decreases when compared to the one used in SVAR estimation (578 stock-days), leaving us with 455 stock-days.

HFTs are in the direction of permanent price movements (with the effects ranging from 0.699 to 4.165 bps per 1-unit increase in various directional activity measures). Trades of Non-Active HFTs have a larger permanent price impact (4.165 bps per additional buyer-initiated trade), while the impact of trades of Active HFTs is smaller at 3.721 bps.²⁹ The impact of aggressive messages on the permanent price dynamics exhibits the opposite pattern with Active HFTs contributing more (0.932 bps) in the direction of permanent price movements than Non-Active HFTs (0.699 bps), though this effect is much smaller in magnitude as compared to trades.

The effect of the messages on pricing error dynamics shows that Non-Active HFTs mitigate pricing errors both through trades (-1.942 bps) and aggressive messages (-0.447 bps).³⁰ At the same time, aggressive messages from Active HFTs are in the same direction as the transitory price movement (0.185 bps). This is consistent with inventory management by market makers (e.g., [Stoll \(1978\)](#) and [Hendershott and Menkveld \(2014\)](#)), as a result of which return reversal is generated.

Table 11 Panels B and D compare contributions to the permanent price component (γ) and transitory price component (δ) from Eq. (12) for Active and Non-Active HFTs in the following regression framework:

²⁹These results are qualitatively similar to [Brogaard, Hendershott, and Riordan \(2014\)](#), who show that per 10,000 USD demanded by HFTs (equivalent of our “trades”), there is a 0.22 bps increase in the permanent price component for large stocks.

³⁰These results are qualitatively similar to [Brogaard, Hendershott, and Riordan \(2014\)](#), who show that per 10,000 USD demanded by HFTs (equivalent of our “trades”), there is a 0.62 bps decrease in the transitory price component for large stocks.

$$\begin{aligned}
Y_{c,s,d}^{imp} = & \eta_s + \xi_d + \\
& + Active_{c,s,d} + \\
& + \beta_1 Return\ Reversal_{s,d} + \beta_2 |Overnight\ Return_{s,d-1 \rightarrow d}| + \\
& + \beta_3 Earnings\ Announcement_{s,d-1 \rightarrow d} + \\
& + \beta_4 Range_{s,d-1} + \beta_5 Closing\ Quoted\ Spread_{s,d-1} + \\
& + \epsilon_{c,s,d}
\end{aligned} \tag{13}$$

where $Y_{c,s,d}^{imp}$ is $\gamma_{c,s,d}^{imp}$ ($\delta_{c,s,d}^{imp}$) of directional activity (*imp*) of stock s , day d , and trader category c . Directional activity measures include aggressive messages from Active and Non-Active HFTs and trades initiated by Active and Non-Active HFTs. We regress $\gamma_{c,s,d}^{imp}$ ($\delta_{c,s,d}^{imp}$) on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. We also include stock and day fixed effects and cluster standard errors by stock.

Panel B of Table 11 shows that aggressive messages (trades) of Active HFTs are systematically contributing more (less) to the permanent price component than Non-Active HFTs by 0.234 bps (-0.444 bps). This is fully in line with the analysis presented in Section 5.1.1 based on the SVAR approach which implicitly models the dynamics of the permanent price component.

Panel D of Table 11 shows that aggressive messages of Active HFTs are systematically contributing more towards pricing errors than those of Non-Active HFTs by 0.632 bps. In addition, though both Active and Non-Active HFTs trade to mitigate pricing errors, the latter have a larger effect by 0.508 bps.

To conclude, we show that both Active and Non-Active HFTs operate in the direction

of permanent price movements, with the latter having a larger price impact through the trades they initiate. Non-Active HFTs help to mitigate pricing errors, while Active HFTs contribute to temporal deviation from the fundamental value through quote adjustments as a result of their inventory management. Overall, this alternative approach to measuring price discovery also supports market-making as preferred strategy of Active HFTs and thus, points in favor of Hypothesis 2 over Hypothesis 1.

6.2 Subsample analysis

In this section, we repeat the analysis for the first half (April 1 – April 15) and the second half (April 16 – April 30) of April 2013. In particular, we repeat the analysis that maps HFTs’ heterogeneity to their decision to participate in the pre-opening period (as in Eq. (2), Eq. (4), and Eq. (5)) and also link this decision to market liquidity measures (as in Eq. (10)) in the two subsamples. To conserve space, we report only the coefficients of the $Active_{c,s,d}$. All the regressions are estimated with stock and day fixed effects and usual control variables.³¹ Standard errors are clustered by stock. The results are reported in Table 12. Overall, the documented effects are consistent between the whole sample, the first half, and the second half of April 2013.

INSERT TABLE 12 HERE

7 Conclusion

In this paper, we attempt to answer the question which type of HFTs chooses to participate in the pre-opening period of the TSE. Our data that identify individual traders across the pre-opening and continuous trading periods allow us to investigate who these traders are, why they decide to come to the pre-opening period, and whether this decision is associated with improved or deteriorated market liquidity.

³¹For the first half of April we cannot include overnight earnings announcements, since they are generally concentrated in the second half of April.

We show that, overall, Active HFTs are market-makers, while Non-Active HFTs are opportunistic traders. Securing time priority in the limit order queue partially explains the participation decision, since orders left unexecuted at the auction enter the continuous trading period with a superior time priority to those submitted after market opens. On the contrary, none of the testable implications of the information channel we examine finds support for it. We also find that decision to participate in the pre-opening period is positively associated with market liquidity in the continuous trading period that follows.

Our findings may be of interest for market design, though we stress that they are only suggestive, since our evidence is associational and the counterfactual liquidity measures are computed under the assumption that other market participants continue to act in the same manner as otherwise. The TSE combines the absence of time priority in the call auction with full time priority for limit orders carried into the continuous trading period, an arrangement shared by most exchanges around the world. This combination is associated with early order placement by market-making HFTs, in a manner consistent with the pre-opening period serving as a queue-formation mechanism for the continuous trading period that follows. Therefore, voluntary participation by HFTs in the pre-opening period is not detrimental to market liquidity and may actually improve it. From a regulatory perspective, voluntary participation by HFTs in the pre-opening period should not be discouraged, since their decision to participate in the pre-opening period is not driven by informational reasons and since we find no evidence of opportunistic behavior by Active HFTs during the continuous trading period that follows.

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Table 1 Summary of traders' characteristics

This table shows summary statistics for the classification of traders during the continuous trading period based on their speed and inventory (HFTs vs. Non-HFTs) and participation during the pre-opening period (Active vs. Non-Active). Traders are classified into four categories on each stock-day with the classification of HFTs and Non-HFTs being permanent, and the Active and Non-Active designations varying from one stock-day to another. First, we compute the speed and inventory measures for each trader-stock-day. Second, we compute the median speed and inventory measures for each trader category, stock, and date. Finally, we compute the median across stock-days for each trader category. Panel A shows the median of the 1%, 5%, and 10% percentiles and the median minimum speed (used to classify traders). Panel B shows the median number of traders, median end-of-day inventory (used to classify traders), median average intraday inventory and its volatility, and the median number of times during the day that the inventory crosses zero. These characteristics are presented per group for April 2013 for 97 stocks from the TOPIX100. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

Panel A: Speed						
		Median of				
		# of traders	Min speed	P1 speed	P5 speed	P10 speed
HFTs	Active	2	0.001	0.002	0.004	0.011
	Non-Active	6	0.001	0.002	0.003	0.005
Non-HFTs	Active	183	4.993	4.995	9.973	20.357
	Non-Active	371	1.018	1.024	2.281	7.089
Panel B: Inventory						
		Median of				
		# of traders	End-of-day inventory	Intraday inventory	Volatility of intraday inventory	# of zero crossings
HFTs	Active	2	1.6%	2.3%	1.9%	25
	Non-Active	6	5.1%	7.0%	4.9%	5
Non-HFTs	Active	183	57.3%	33.4%	19.1%	0
	Non-Active	371	75.0%	44.2%	24.5%	0

Table 2 Structural impulse response functions

This table presents the analysis of the structural impulse response functions (SIRFs) from the structural vector-autoregression (SVAR) approach as defined in Eq. (1). SVAR models are estimated in event time, where the event can be any message submitted. SIRFs are computed for returns and cumulated over 20 steps ahead for a +1 shock to directional activity. Directional activity measures include aggressive messages from Active and Non-Active HFTs and trades initiated by Active and Non-Active HFTs. Panel A reports average SIRFs across stock-days for Active and Non-Active HFTs, respectively. Panel B compares SIRFs of Active and Non-Active HFTs in the morning continuous period using a regression framework. In particular, we regress SIRFs on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d . Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

Panel A: Structural impulse response functions						
	Aggressive messages			Trades		
	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference
Pre-opening period	0.233***	—	—	—	—	—
Morning continuous period	1.282***	0.555***	0.727***	4.434***	6.261***	−1.826***
Panel B: Difference in SIRFs of Active and Non-Active HFTs						
	Aggressive messages			Trades		
	(1)	(2)	(3)	(4)	(5)	(6)
$Active_{c,s,d}$	0.727*** (15.022)	0.727*** (15.022)	0.727*** (14.989)	−1.826*** (−15.263)	−1.826*** (−15.263)	−1.826*** (−15.230)
$Return\ Reversal_{s,d}$			−0.003 (−0.195)			0.043 (0.704)
$ Overnight\ Return _{s,d-1 \rightarrow d}$			0.009 (0.554)			−0.009 (−0.132)
$Earnings\ Announcement_{s,d-1 \rightarrow d}$			−0.074** (−2.004)			−0.064 (−0.125)
$Range_{s,d-1}$			−0.010 (−1.044)			−0.079* (−1.699)
$Closing\ Quoted\ Spread_{s,d-1}$			0.015 (0.165)			1.427** (2.615)
$Constant$	0.555*** (17.629)			6.261*** (20.466)		
Stock FE	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	1,156	1,156	1,156	1,156	1,156	1,156
Adjusted R^2	0.379	0.715	0.714	0.161	0.767	0.769

Table 3 Trading volume imbalance

This table presents an analysis of trading volume imbalance ratio during the morning continuous trading period for Active and Non-Active HFTs. First, we identify whether a trader initiated a transaction (demand liquidity) or not (supply liquidity) and compute for each trader the imbalance ratio between supplied minus demanded volume relative to total trading volume. Second, we compute trading volume imbalance (in %) per trader category, stock, and date as the average of the aforementioned ratio. We use both trading volume in JPY and number of shares. We regress the trading volume imbalance on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

Panel A: Trading volume imbalance						
	Based on JPY volume			Based on share volume		
	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference
Morning continuous period	67.331***	-2.670	70.001***	67.329***	-2.670	69.999***
Panel B: Difference in trading volume imbalance of Active and Non-Active HFTs						
	Based on JPY volume			Based on share volume		
	(1)	(2)	(3)	(4)	(5)	(6)
$Active_{c,s,d}$	70.001*** (20.798)	70.001*** (20.798)	70.001*** (20.753)	69.999*** (20.797)	69.999*** (20.797)	69.999*** (20.752)
$Return\ Reversal_{s,d}$			2.014** (2.127)			2.016** (2.129)
$ Overnight\ Return _{s,d-1 \rightarrow d}$			0.727 (0.859)			0.725 (0.858)
$Earnings\ Announcement_{s,d-1 \rightarrow d}$			14.058*** (4.700)			14.064*** (4.687)
$Range_{s,d-1}$			1.787*** (3.114)			1.787*** (3.113)
$Closing\ Quoted\ Spread_{s,d-1}$			2.505 (0.370)			2.505 (0.371)
$Constant$	-2.670 (-0.849)			-2.670 (-0.849)		
Stock FE	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	1,156	1,156	1,156	1,156	1,156	1,156
Adjusted R^2	0.670	0.749	0.750	0.670	0.749	0.750

Table 4 Presence in the limit order book

This table presents an analysis of liquidity provision during the morning continuous trading period for Active and Non-Active HFTs. First, we compute for each trader passive presence at either side of the limit order book (as a percentage of total time) for the top, best bid-offer level (BBO), and three best levels (3BBO) of the limit order book. Second, we compute an average passive presence and depth for each trader category, stock, and date. Panel A reports average presence across stock-days for Active and Non-Active HFTs, respectively. Panel B compares the presence of Active and Non-Active HFTs in the morning continuous period using a regression framework. We regress passive presence and depth on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d . Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

Panel A: Passive presence									
	Top			BBO			3BBO		
	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference
Morning continuous period	8.263***	5.186***	3.077***	34.457***	28.478***	5.979***	57.013***	56.460***	0.553
Panel B: Difference in passive presence of Active and Non-Active HFTs									
	Top			BBO			3BBO		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Active_{c,s,d}$	3.077*** (4.848)	3.077*** (4.848)	3.077*** (4.837)	5.979*** (3.542)	5.979*** (3.542)	5.979*** (3.534)	0.553 (0.276)	0.553 (0.276)	0.553 (0.275)
$Return\ Reversal_{s,d}$			0.368 (1.556)			0.328 (0.726)			0.481 (1.018)
$ Overnight\ Return _{s,d-1 \rightarrow d}$			0.628** (2.531)			0.565 (0.998)			0.311 (0.458)
$Earnings\ Announcement_{s,d-1 \rightarrow d}$			-3.537* (-1.801)			-5.919** (-2.165)			-0.445 (-0.247)
$Range_{s,d-1}$			0.072 (0.456)			-0.551* (-1.698)			0.083 (0.214)
$Closing\ Quoted\ Spread_{s,d-1}$			0.820 (0.488)			7.540** (2.198)			5.622 (1.334)
$Constant$	5.186*** (12.976)			28.478*** (20.936)			56.460*** (76.190)		
Stock FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	1,156	1,156	1,156	1,156	1,156	1,156	1,156	1,156	1,156
Adjusted R^2	0.070	0.223	0.224	0.040	0.400	0.400	-0.001	0.163	0.160

Table 5 HFT's decision to participate in the pre-opening period

This table reports the determinants of the decision to participate in the pre-opening period that HFTs make (see Eq. (6)). Specifications (1)–(3) use a dummy variable that takes value of one, if HFT k is active in the pre-opening period for stock s on day d , and zero otherwise, as the dependent variable. Specifications (4)–(6) use the % of total number of messages in the pre-opening period that come from HFT k as the dependent variable. Specifications (7)–(9) use a dummy variable that takes value of one, if HFT k switches from Non-Active to Active HFT for stock s on day d , and zero otherwise, as the dependent variable. Independent variables are a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, the previous day stock liquidity proxied by the closing bid-ask spread, and the absolute value of cumulative end-of-day inventory (in 100 mln JPY). Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

	Active in the pre-opening period			% of messages in the pre-opening period			Switching from Non-Active to Active HFT		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Return Reversal</i> _{s,d}	−0.005 (−0.808)	−0.005 (−1.353)	−0.007* (−1.730)	0.001 (0.051)	−0.003 (−0.246)	−0.007 (−0.637)	−0.005 (−0.948)	−0.005 (−1.131)	−0.006 (−1.425)
<i> Overnight Return </i> _{$s,d-1 \rightarrow d$}	−0.020*** (−3.761)	−0.017*** (−4.071)	−0.018*** (−4.401)	−0.068*** (−4.038)	−0.027*** (−2.889)	−0.028*** (−2.946)	−0.018*** (−4.845)	−0.013*** (−2.813)	−0.016*** (−3.859)
<i>Earnings Announcement</i> _{$s,d-1 \rightarrow d$}	−0.041 (−0.949)	0.016 (0.677)	0.006 (0.220)	0.030 (0.427)	−0.042 (−0.729)	−0.049 (−0.989)	−0.018 (−0.401)	0.033* (1.735)	0.005 (0.212)
<i>Range</i> _{$s,d-1$}	0.008* (1.732)	0.003 (0.819)	0.002 (0.689)	0.017 (0.942)	0.003 (0.392)	0.003 (0.408)	0.003 (1.128)	−0.0004 (−0.110)	0.0001 (0.041)
<i>Closing Quoted Spread</i> _{$s,d-1$}	0.319*** (6.808)	0.007 (0.274)	−0.004 (−0.141)	0.365 (1.432)	0.202*** (2.746)	0.180** (2.372)	0.101*** (4.204)	−0.009 (−0.323)	−0.019 (−0.768)
<i> Cumulative EOD Inventory </i> _{$k,s,d-1$}	0.002 (0.366)	−0.032*** (−5.056)	−0.001 (−0.184)	−0.027* (−1.787)	−0.030* (−1.806)	−0.003 (−0.167)	0.001 (0.971)	−0.015*** (−7.779)	−0.001 (−0.801)
Trader FE	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes
Stock FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	12,633	12,633	12,633	12,633	12,633	12,633	9,323	9,323	9,323
Adjusted R^2	0.378	0.090	0.434	0.277	0.064	0.332	0.306	0.058	0.350

Table 6 Profits

This table presents analysis of relative profits made by Active and Non-Active HFTs during 09:00–09:15, 09:00–09:30, and 09:00–09:45. First, we compute profits for each trader-stock-day under the assumption that the trader starts every stock-day with a zero inventory position as in Eq. (7). Second, we compute an average relative profit for each trader category, stock, and date. Panel A reports average relative profits across stock-days for Active and Non-Active HFTs, respectively. Panel B compares relative profits of Active and Non-Active HFTs in the morning continuous period using a regression framework. We regress relative profit on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d . Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

Panel A: Relative profits									
	09:00–09:15			09:00–09:30			09:00–09:45		
	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference
Morning continuous period	2.101***	2.744***	−0.643	1.255**	1.395**	−0.140	0.958**	1.882***	−0.925
Panel B: Difference in relative profits of Active and Non-Active HFTs									
	09:00–09:15			09:00–09:30			09:00–09:45		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Active_{c,s,d}$	−0.643 (−0.756)	−0.643 (−0.756)	−0.643 (−0.755)	−0.140 (−0.179)	−0.140 (−0.179)	−0.140 (−0.178)	−0.925 (−1.227)	−0.925 (−1.227)	−0.925 (−1.224)
$Return\ Reversal_{s,d}$			0.310 (0.388)			0.234 (0.331)			0.425 (0.605)
$ Overnight\ Return _{s,d-1 \rightarrow d}$			−0.837 (−0.999)			−0.947 (−1.090)			−0.710 (−0.994)
$Earnings\ Announcement_{s,d-1 \rightarrow d}$			−2.391 (−0.350)			−7.597*** (−2.989)			0.005 (0.003)
$Range_{s,d-1}$			−0.706 (−0.749)			−0.200 (−0.300)			−0.431 (−0.709)
$Closing\ Quoted\ Spread_{s,d-1}$			−0.485 (−0.087)			−4.200 (−0.777)			−2.003 (−0.376)
<i>Constant</i>	2.744*** (4.059)			1.395** (2.526)			1.882*** (3.340)		
Stock FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	1,156	1,156	1,156	1,156	1,156	1,156	1,156	1,156	1,156
Adjusted R^2	−0.0003	0.050	0.048	−0.001	0.042	0.041	0.001	0.046	0.044

Table 7 Presence in the limit order book: Role of early limit orders

This table shows the relation between liquidity provision during the morning continuous trading period and the proportion of orders entered in the pre-opening period but executed during the continuous trading period for Active HFTs only. For each trader, we compute passive presence on either side of the limit order book (as a percentage of total time) for the top, best bid-offer level (BBO), and three best levels (3BBO) of the limit order book. We compute an average passive presence and depth for each trader category, stock, and date. We regress passive presence and depth on the proportion of orders that HFTs enter during the pre-opening period that get executed in the continuous trading period. Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

	Top	BBO	3BBO
$EarlyOrder_{ActiveHFT,s,d}$	0.012 (0.260)	0.270*** (3.260)	0.332*** (3.135)
$Return\ Reversal_{s,d}$	1.173*** (3.461)	0.345 (0.538)	0.015 (0.016)
$ Overnight\ Return _{s,d-1 \rightarrow d}$	0.862** (2.232)	1.542 (1.469)	1.929 (1.407)
$Earnings\ Announcement_{s,d-1 \rightarrow d}$	-6.687* (-1.928)	-13.736*** (-2.813)	-5.826* (-1.754)
$Range_{s,d-1}$	0.252 (1.033)	-0.135 (-0.218)	0.891 (1.103)
$Closing\ Quoted\ Spread_{s,d-1}$	5.579** (2.045)	16.152*** (3.781)	22.490*** (3.446)
Stock FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock
Observations	578	578	578
Adjusted R^2	0.399	0.684	0.642

Table 8 Market liquidity

This table shows an analysis of the relation between market liquidity as proxied by proportional quoted spread (in bps), maximum quoted depth between best bid and offer (in mln JPY), and depth imbalance between best bid and offer (in %) and Active HFTs (all measures are computed right before transaction takes place and aggregated to a stock-date level using share-weighted average). We compute these measures using all the data and also the data without limit orders stemming from Active HFTs (under the assumption that other market participants continue to act in the same manner). We regress market liquidity measures on a dummy variable that takes value of one, if limit orders from Active HFTs are included in the data, and zero otherwise for stock s and day d ($Active_{c,s,d}$). Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, and the previous day stock volatility proxied by high minus low range. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

	Proportional quoted spread (in bps)			Depth (in mln JPY)			JPY depth imbalance (in %)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Active_{c,s,d}$	-0.222*** (-11.605)	-0.222*** (-11.605)	-0.222*** (-11.585)	0.650*** (9.417)	0.650*** (9.417)	0.650*** (9.400)	-0.076** (-2.109)	-0.076** (-2.109)	-0.076** (-2.105)
$Return\ Reversal_{s,d}$			0.046 (0.559)			-2.425*** (-3.075)			-0.658 (-1.128)
$ Overnight\ Return _{s,d-1 \rightarrow d}$			0.015 (0.191)			0.494 (0.609)			-0.106 (-0.212)
$Earnings\ Announcement_{s,d-1 \rightarrow d}$			0.407 (0.654)			4.002 (0.348)			-7.718* (-1.859)
$Range_{s,d-1}$			-0.062 (-0.566)			-2.084*** (-3.362)			-0.342 (-0.906)
$Constant$	14.538*** (20.313)			34.623*** (13.082)			1.070*** (3.227)		
Stock FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	1,156	1,156	1,156	1,156	1,156	1,156	1,156	1,156	1,156
Adjusted R^2	-0.0004	0.945	0.945	-0.001	0.794	0.798	-0.001	0.267	0.271

Table 9 Market liquidity: Role of early limit orders

This table shows the relation between the difference in the market liquidity with and without orders from Active HFTs during the morning continuous trading period and the proportion of orders entered in the pre-opening period but executed during continuous trading period for Active HFTs only. Market liquidity as proxied by the proportional quoted spread (in bps), the maximum quoted depth between the best bid and offer (in mln JPY), and the depth imbalance between the best bid and offer (in %) and Active HFTs (all measures are computed right before transaction takes place and aggregated to a stock-date level using share-weighted average). We regress these measures on the proportion of orders that HFTs enter during the pre-opening period that get executed in the continuous trading period. Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, and the previous day stock volatility proxied by high minus low range. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

	Proportional quoted spread (in bps)	Depth (in mln JPY)	JPY depth imbalance (in %)
	(1)	(2)	(3)
<i>% of Early Orders_{ActiveHFT,s,d}</i>	−0.0004 (−0.516)	0.009*** (2.812)	−0.011** (−2.217)
<i>Return Reversal_{s,d}</i>	−0.005 (−0.575)	−0.044 (−1.058)	0.125* (1.705)
<i> Overnight Return _{s,d−1→d}</i>	−0.007 (−0.882)	0.015 (0.403)	0.008 (0.124)
<i>EarningsAnnouncement_{t,s,d−1→d}</i>	−0.102*** (−4.015)	−0.017 (−0.150)	−1.233** (−2.480)
<i>Range_{s,d−1}</i>	−0.017* (−1.839)	−0.024 (−1.106)	0.071 (1.412)
Stock FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock
Observations	578	578	578
Adjusted R^2	0.590	0.557	0.076

Table 10 Extreme price movements

This table shows the results of extreme price movement (EPM) analysis following [Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov \(2018\)](#). EPMs are defined as 99.9th percentile of absolute 10-sec midquote returns. Supply (demand) is defined as the difference in the number of shares supplied (demanded) in the direction vs. the opposite direction of EPM. Net share imbalance is a sum of supply and demand metrics. Supply, demand, and net share imbalance are computed for Active and Non-Active HFTs, respectively, for a window of $-/+$ 20 seconds around EPM. We report cumulative average returns and average supply, demand, and net share imbalance across both positive and negative EPMs (metrics for negative EPMs are inverted). t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

	-20	-10	0	+10	+20
Panel A: Cumulative return					
	-0.638	3.146	60.515	63.072	63.625
Panel B: Active HFTs					
Net	-73.972 (-0.867)	-114.604 (-0.863)	-1120.356*** (-3.891)	35.360 (0.298)	121.831 (0.791)
Demand	-32.497 (-0.913)	-52.282 (-0.854)	120.720 (0.877)	-92.748** (-2.019)	38.435 (0.806)
Supply	-41.475 (-0.552)	-62.322 (-0.664)	-1241.076*** (-4.381)	128.109 (1.141)	83.396 (0.533)
Panel C: Non-Active HFTs					
Net	510.346 (0.916)	845.674 (1.270)	14977.163*** (8.308)	646.525 (1.204)	-600.958 (-1.347)
Demand	362.441 (0.687)	793.457 (1.298)	15971.714*** (9.187)	357.668 (0.696)	-559.249 (-1.386)
Supply	147.905 (1.522)	52.217 (0.397)	-994.551*** (-4.573)	288.857*** (3.621)	-41.708 (-0.326)

Table 11 State-space approach

This table presents the analysis of the state-space model for price dynamics, as defined in Eq. (12). State-space models are estimated for the morning continuous trading period only, using the Kalman filter. State-space models are estimated at event time, where the event can be any message with the potential to impact the midquote. Directional activity measures include aggressive messages from Active and Non-Active HFTs and trades initiated by Active and Non-Active HFTs. Panel A (Panel C) reports the average contribution to the permanent (transitory) price component: γ (δ). Panel B (Panel D) compares the contribution to the permanent (transitory) price component of Active and Non-Active HFTs in the morning continuous period using a regression framework. In particular, we regress contribution to the permanent (transitory) price component on a dummy variable that takes value of one, if trader category c is equal to Active HFTs, and zero otherwise for stock s and day d . Control variables include a dummy variable for return reversal, the absolute value of overnight return, a dummy variable for overnight earnings announcement, the previous day stock volatility proxied by high minus low range, and the previous day stock liquidity proxied by the closing bid-ask spread. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

Panel A: Contribution to permanent price component (γ)						
	Aggressive messages			Trades		
	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference
Morning continuous period	0.932***	0.699***	0.234***	3.721***	4.165***	-0.444***
Panel B: Difference in γ of Active and Non-Active HFTs						
	Aggressive messages			Trades		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Active</i> _{c,s,d}	0.234*** (3.925)	0.234*** (3.925)	0.234*** (3.914)	-0.444*** (-2.766)	-0.444*** (-2.766)	-0.444*** (-2.758)
<i>Return Reversal</i> _{s,d}			-0.092 (-1.550)			0.211 (1.399)
<i> Overnight Return </i> _{$s,d-1 \rightarrow d$}			0.036 (0.981)			0.077 (0.910)
<i>Earnings Announcement</i> _{$s,d-1 \rightarrow d$}			0.433 (1.420)			1.498*** (3.784)
<i>Range</i> _{$s,d-1$}			-0.009 (-0.299)			-0.261** (-2.250)
<i>Closing Quoted Spread</i> _{$s,d-1$}			-0.099 (-0.363)			1.772** (2.299)
<i>Constant</i>	0.699*** (18.288)			4.165*** (15.260)		
Stock FE	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	910	910	910	910	910	910
Adjusted R^2	0.022	0.159	0.160	0.006	0.384	0.389

Table 11 State-space approach (continued)

Panel C: Contribution to transitory price component (δ)						
	Aggressive messages			Trades		
	Active HFTs	Non-Active HFTs	Difference	Active HFTs	Non-Active HFTs	Difference
Morning continuous period	0.185***	-0.447***	0.632***	-1.435***	-1.942***	0.508***
Panel D: Difference in δ of Active and Non-Active HFTs						
	Aggressive messages			Trades		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Active_{c,s,d}</i>	0.632*** (8.563)	0.632*** (8.563)	0.632*** (8.540)	0.508*** (4.250)	0.508*** (4.250)	0.508*** (4.238)
<i>Return Reversal_{s,d}</i>			0.067 (1.271)			-0.048 (-0.309)
<i> Overnight Return _{s,d-1→d}</i>			-0.034 (-0.832)			-0.040 (-0.373)
<i>Earnings Announcement_{s,d-1→d}</i>			0.077 (0.686)			-0.453 (-0.734)
<i>Range_{s,d-1}</i>			-0.004 (-0.130)			0.140 (1.430)
<i>Closing Quoted Spread_{s,d-1}</i>			0.206 (1.047)			-0.372 (-0.472)
<i>Constant</i>	-0.447*** (-16.502)			-1.942*** (-13.185)		
Stock FE	No	Yes	Yes	No	Yes	Yes
Date FE	No	Yes	Yes	No	Yes	Yes
Clustered SE	by Stock	by Stock	by Stock	by Stock	by Stock	by Stock
Observations	910	910	910	910	910	910
Adjusted R^2	0.191	0.227	0.225	0.015	0.214	0.211

Table 12 Subsample analysis

This table presents subsample analysis for the 1st and 2nd half of April 2013 focusing on the difference between Active and Non-Active HFTs in terms of the SIRFs as in Eq. (2) (Panel A), trading volume imbalance as in Eq. (4) (Panel B), passive presence in the limit order book as in Eq. (5) (Panel C), and contribution to market liquidity as in Eq. (10) (Panel D). To conserve space, we report only the coefficients of $Active_{c,s,d}$. All regressions include usual control variables and stock and day fixed effects. Standard errors are clustered by stock; t -statistics are reported in parentheses; ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Continuous variables are winsorized at the 1% and 99% levels. Our sample includes 97 stocks from the TOPIX100 during April 2013. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

	1st half	2nd half
Panel A: SIRF		
Aggressive messages	0.716*** (13.246)	0.739*** (14.720)
Trades	-1.973*** (-13.670)	-1.661*** (-13.573)
Panel B: Trading volume imbalance		
Trading volume imbalance (JPY)	68.945*** (19.034)	71.198*** (19.937)
Trading volume imbalance (shares)	68.936*** (19.036)	71.203*** (19.932)
Panel C: Passive presence		
Top	2.806*** (4.051)	3.383*** (4.426)
BBO	5.745*** (2.998)	6.245*** (3.365)
3BBO	1.818 (0.812)	-0.881 (-0.358)
Panel D: Market liquidity		
PQSPR	-0.261*** (-10.528)	-0.177*** (-11.639)
Depth (mln JPY)	0.607*** (9.373)	0.698*** (7.780)
JPY Depth Imbalance (%)	-0.033 (-0.732)	-0.125** (-2.184)

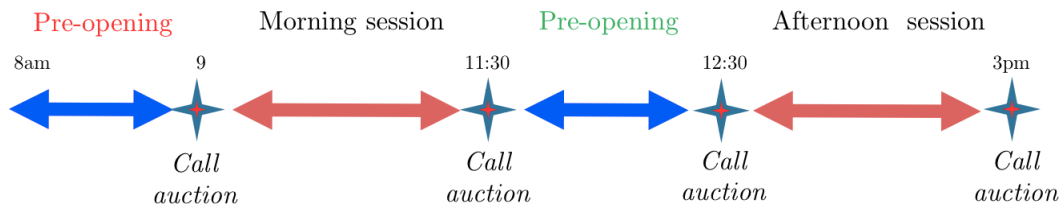


Figure 1. Trading schedule of the Tokyo Stock Exchange (TSE). This figure shows the trading schedule of the TSE on a typical trading day. The two separate sessions (morning and afternoon) start at 9:00 a.m. and 12.30 p.m., respectively. A call auction precedes the subsequent continuous trading period.

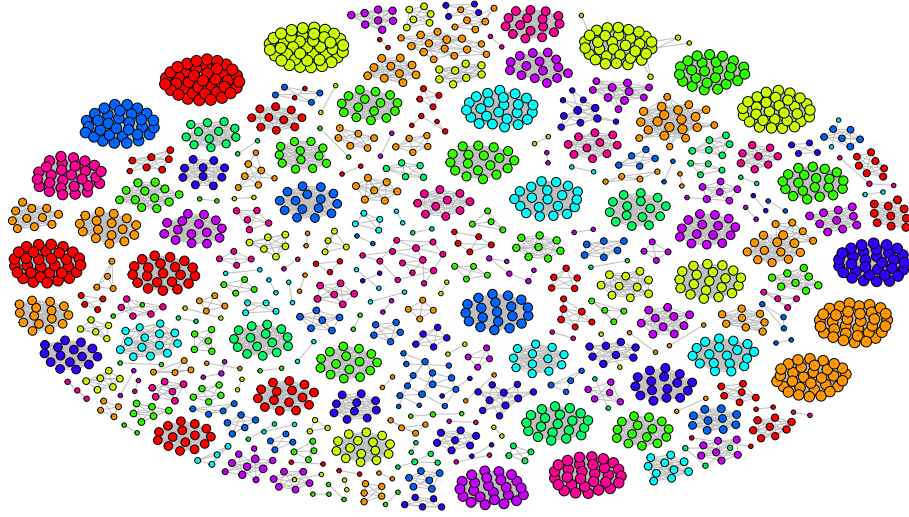


Figure 2. Graphical representation of usage of virtual servers (VSs) by traders. This graph displays the relation between the number of VSs and the number of trading desks during April 2013 on the Tokyo Stock Exchange (TSE) for 97 stocks from the TOPIX100. The total number of VSs is 3,661 (all the dots in the figure), while the number of trading desks using one or more virtual servers is 2,128 (the colored groups in the figure). Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

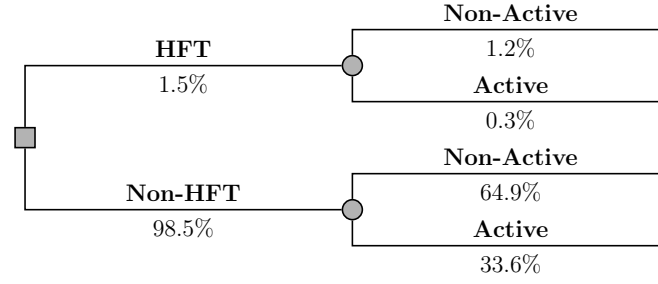


Figure 3. Trader classification. This diagram shows the average proportion of traders that fall into one of the categories used in the paper. More specifically, we classify traders based on their speed and inventory as HFTs vs. Non-HFTs and based on their participation in the pre-opening period as Active vs. Non-Active, with the classification of HFTs and Non-HFTs being permanent and the Active and Non-Active designations varying from one stock-day to another. These proportions are presented for April 2013, for 97 stocks from the TOPIX100. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

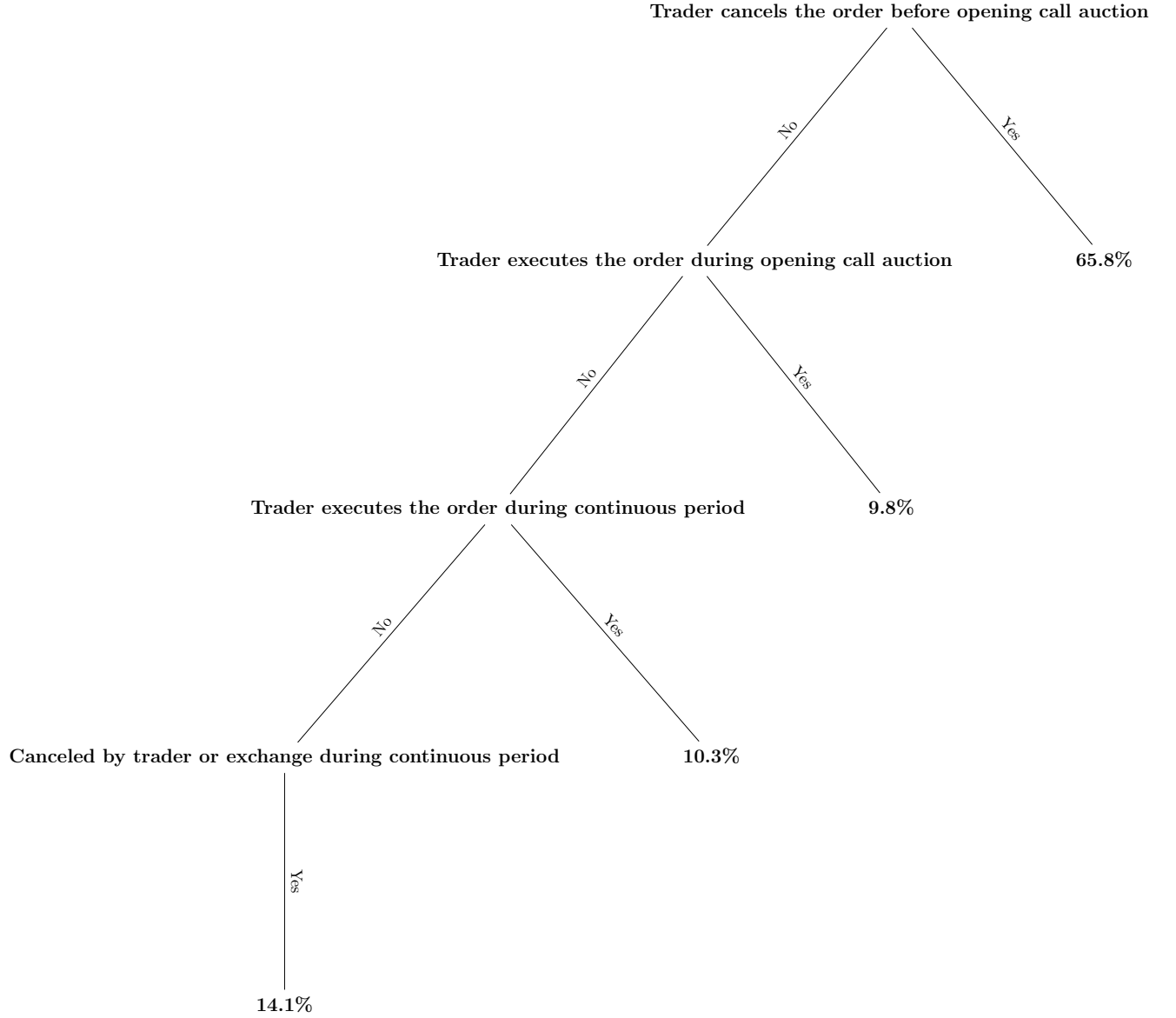


Figure 4. Orders entered by HFTs during pre-opening period. This figure shows decisions that HFTs make regarding the orders they submit during the pre-opening period. We report proportions of unique orders averaged across stock-days that were canceled during the pre-opening period, orders that were executed in the opening auction, orders that were executed during the morning continuous period, and orders that were canceled during continuous period either by trader or stock exchange. Our sample includes 97 stocks from the TOPIX100. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

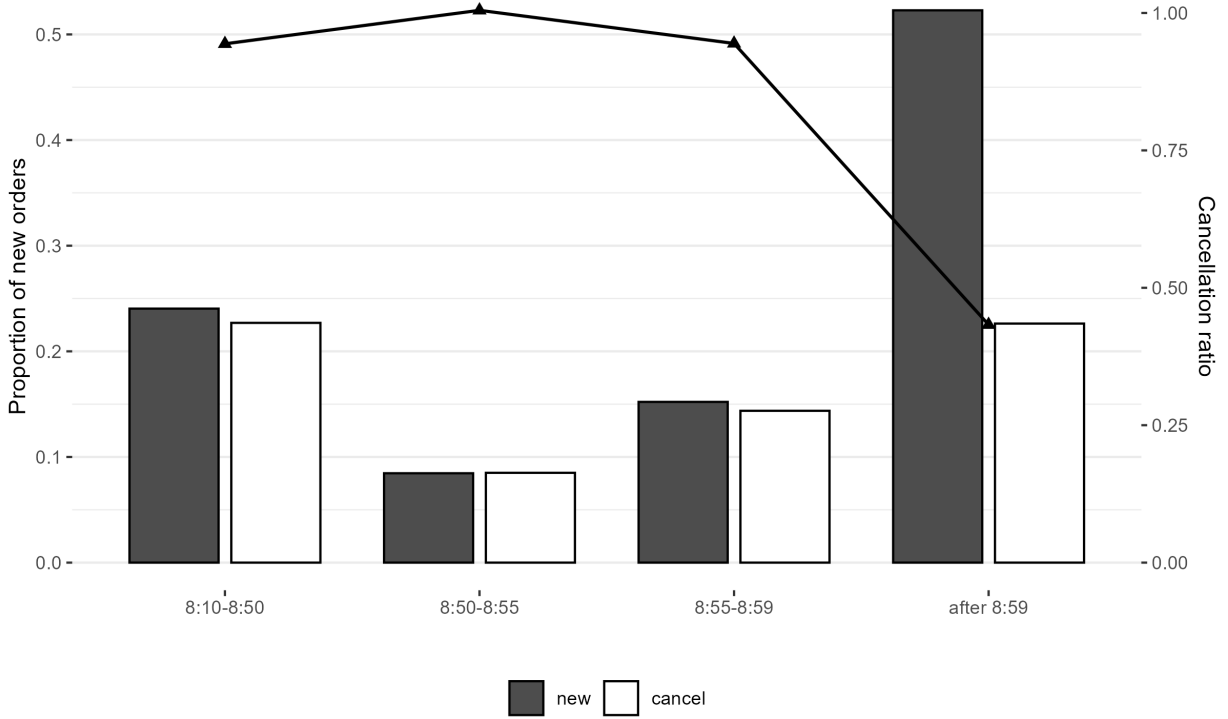


Figure 5. New orders and cancellations by HFTs during the pre-opening period. This figure shows new orders and cancellations stemming from HFTs during 8:10-8:50, 8:50-8:55, 8:55-8:59, and after 8:59 till the opening auction takes place. The new orders and cancellations per sub-period are reported on the left axis as a proportion of new orders submitted during the pre-opening period and averaged across stock-days. Cancellation ratio per sub-period averaged across stock-days is reported on the right axis. Our sample includes 97 stocks from the TOPIX100. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.

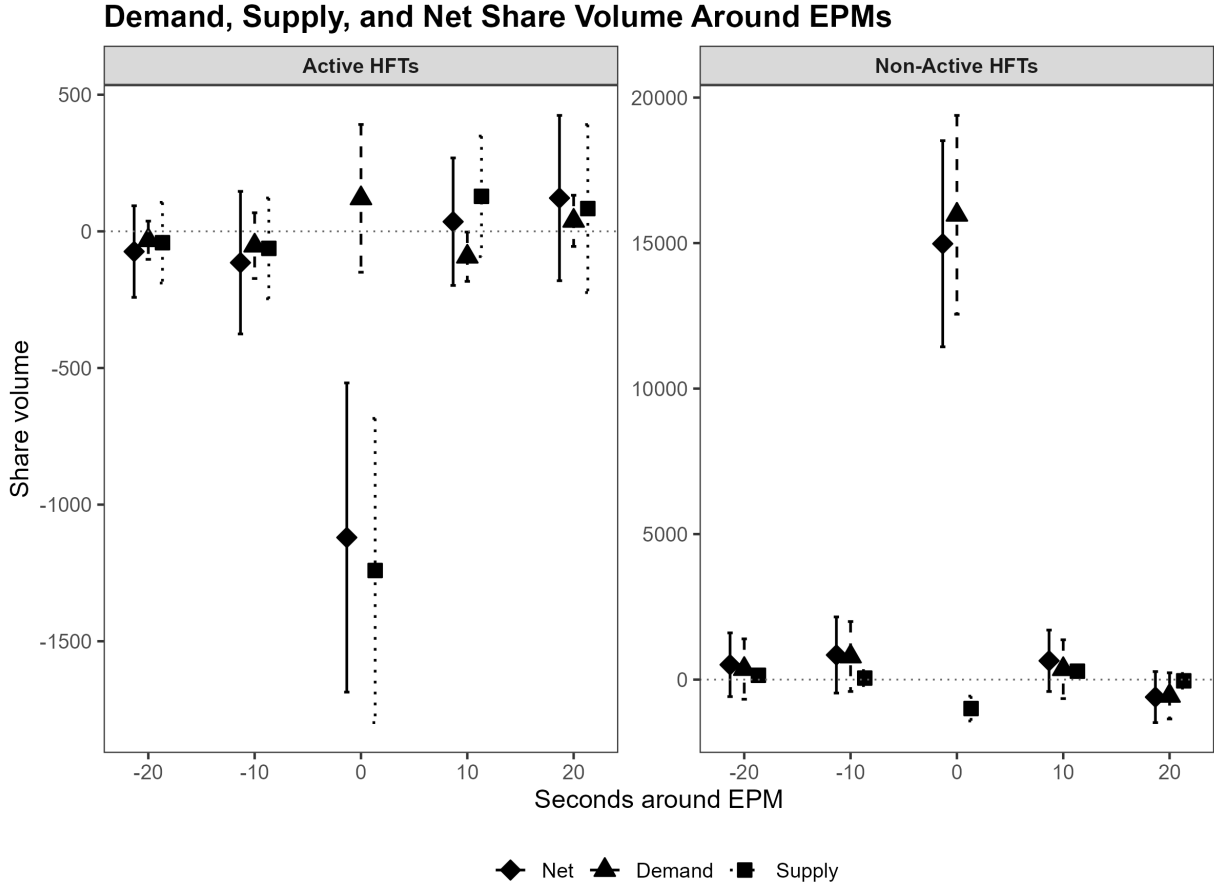


Figure 6. Activity of HFTs around extreme price movements. This figure shows extreme price movement (EPMs) and trading activity of Active and Non-Active HFTs. EPMs are defined as 99.9th percentile of absolute 10-sec midquote returns. Supply (demand) is defined as the difference in the number of shares supplied (demanded) in the direction vs. the opposite direction of EPM. Net share imbalance is a sum of supply and demand metrics. Supply, demand, and net share imbalance are computed for Active and Non-Active HFTs, respectively, for a window of $-/+$ 20 seconds around EPM. Our sample includes 97 stocks from the TOPIX100. Order flow data with order IDs and virtual server IDs were provided by the Tokyo Stock Exchange.